ZOOLOGY NOTES.

- Worthy for CSS, PMS, PPSC, FPSC
- Following topics of CSS are covered in these notes.

- **Cell concept and cell theory, Organization of cellular organelle (their structure and functions), Central dogma of cell biology (Transcription and Translation), Meiosis and Mitosis**
- **The chemical basis of animal life: Brief introduction to bio-molecules; carbohydrates, lipids, proteins and nucleic acids.**
- **Tissues Types: epithelial, connective, muscle and nervous tissue; organs and organ systems.**
- **Enzymes function and factors affecting their activity, cofactors and coenzymes.**
- **Energy Harvesting: Aerobic and anaerobic respiration the major source of ATP.**

- Following topics of PMS are covered in these notes.

- **Generalized Structure of Prokaryotic and Eukaryotic Cell.**
- **Morphology, chemical composition and Functions of cellular organelles.**
- **Enzymes Catalysis, Regulation & Inhibition,**
- **Metabolic Pathways, Glycolysis, Krebs cycle and Electrons Transport chain.**
- **Nucleic acid.**
- **Mechanism of Protein synthesis, Transcription and Translation,**
- **Mitosis, Meiosis.**

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Cell Biology

- What is cell biology?

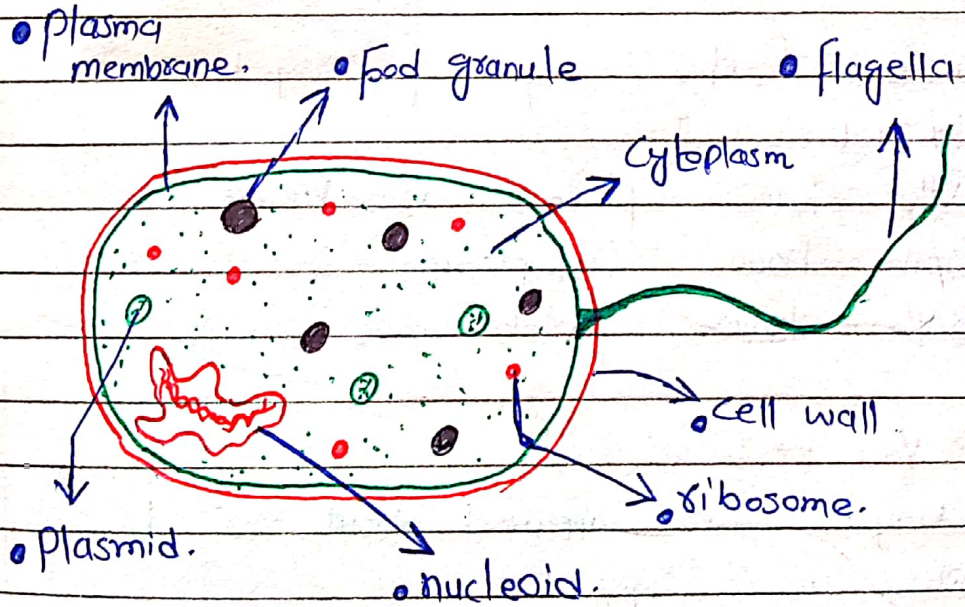
The branch of biology that deals with the structure, function & behaviour of cell.

- (Topic 1)

↳ Generalized Structure of prokaryotic & eukaryotic cell.

- What is prokaryotic Cell?

It is a simple - single Celled (unicellular) organism that lack's cell's membrane bound organelle & nucleus.



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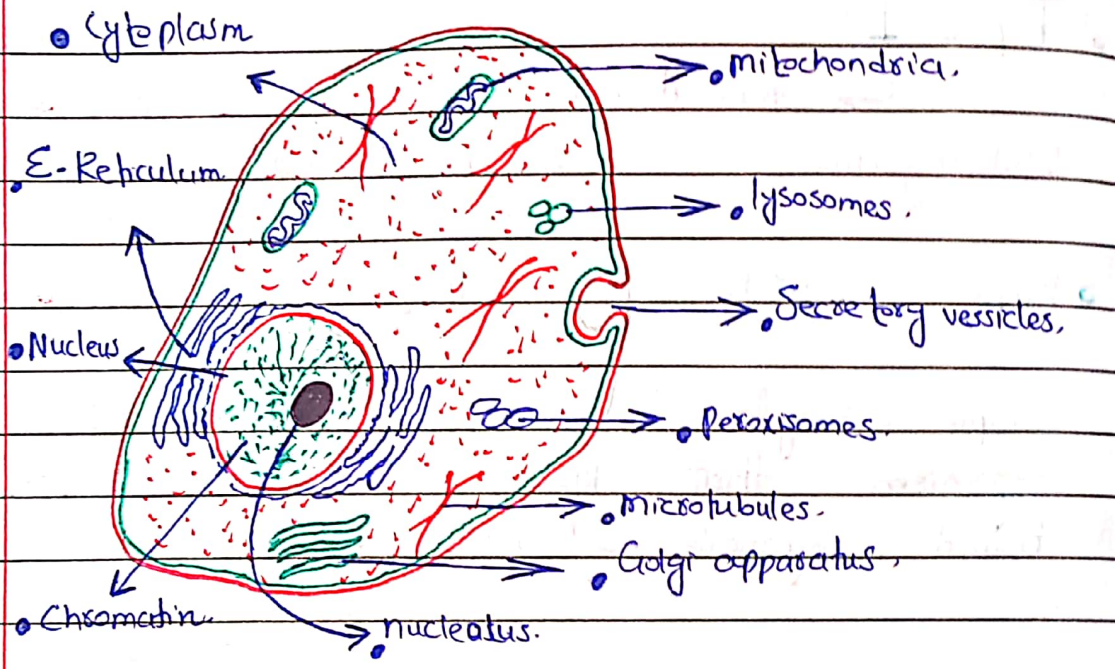
What is Eukaryotic cell?

It is a cell that has a membrane bound nucleus & other membrane bound Compartments or sacs called organelles which have specialized functions.



Common features of prokaryotic & eukaryotic cell.

- Both have plasma membrane.
- Both have Cytoplasm
- Both have ribosomes

Do You Know?

- prokaryotic Cells appeared about 3.5 to 3.8 billion years ago.
- Eukaryotic Cells appeared about 2.7 billion years ago.

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Different features of prokaryotic & eukaryotic cells.

Prokaryotic Cell.	Eukaryotic Cell.
The word prokaryote is derived from a Greek word Pro means before & Karyon means Nucleus.	The word Eukaryote is derived from a Greek word Eu mean's After & Karyon mean's Nucleus.
Smaller in size i.e 1µm to 10µm.	larger in size ie to
Donot have membrane bound organellies.	Contain membrane bound organellies.
Genetic material is not closed by nuclear membrane.	Genetic material is closed by Nuclear membrane.
Histone's are not found in prokaryotic Cells.	Histone's are found in Eukaryotic Cells.
Mitochondria are Absent in prokaryotes.	Mitochondria are present in Eukaryotic Cells.

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• Cell division is Simply by fission No, mitosis & meiosis occure.	• Cell division occure by meiosis.
• RNA Synthesize & proceed's in nucleus same Compartments.	• RNA synthesis in nucleus & proteins Synthesized in Cytoplasm.
• lysosomes are not found.	• lysosome's are fund in Eukaryotes.
• E.g. • Bacteria. • Cynobacteria.	• Eg. plants, Animal Algae, fungi.

• Discribe the Morphology, Chemical Composition & function's of Cellular organellies.

• What is organelle ?

An organelle is a Subcellular Structure that has one or more specific jobs to perform in the Cell, much like an organ does in the body.

Note.

"Cellular organellies are included in Syllabus of both. CSS, PMS."

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(CSS, PMS)

Cytoplasm

Defination.

The living Content of protoplasm between plasma membrane & nucleus is called Cytoplasm.

- Nucleus + Cytoplasm ⟶ Protoplasm

Components of Cytoplasm.

Main Components of Cytoplasm are the following.....

- Cell organellies.
- Storage products
- Insoluble waste
- Cytosd.

(i) Cytosol.

→ It is a mainpart of Cytoplasm & has all materials & structures in it.

→ Chemically it is composed of 80% water.

→ Smaller molecules in Cytosd form true Solution & larger molecules form Colloidal Solution.

Cell organellies.

Various Cellular organellies are Suspended in Cytoplasm.

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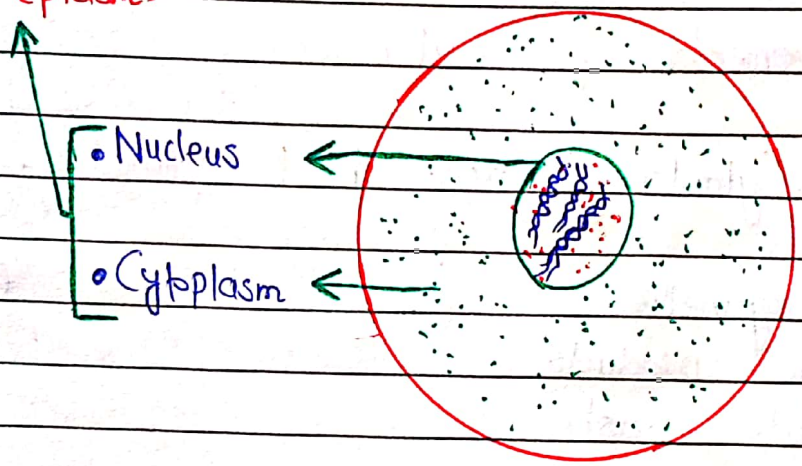
- **Storage products.**

Numarious storage vasicle's are also a part of Cytoplasm.

- **Insoluble wastes.**

It is also a storehouse for various Cellular Wastes.

- **Protoplasm.**



- Eukaryotic cell.

Major functions of Cytoplasm.

- It work's as a store house for various vital chemical's of Cell.
- It also act as site for various metabolic pathways.
- Cytoplasm keep's all organellie's in motion by providing motion path.
- It also form new Cell wall.

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Endoplasmic Reticulum. (CSS, PMS)

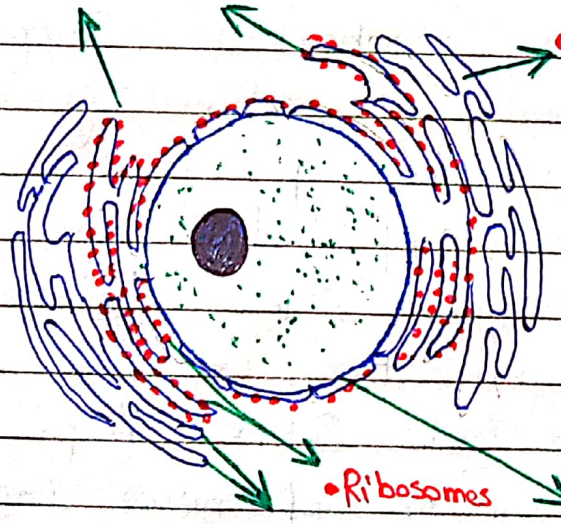
Dafination.

It is a network of membranous tubules within the Cytoplasm of eukaryotic cell Continuous with the nuclear membrane.

Structure.

ER is the Continuous form of nuclear evelope to the plasma membrane. It is a series of channels. These Channels vary in size & shape in different cells. The material present in these ~~cells~~ is separated from the Cytoplasmic materials by flattened tubular membranes called cisternae.

Rough endoplasmic Reticulum.



E. Reticulum.

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Types of Endoplasmic Reticulum.

- **Rough Endoplasmic Reticulum.**

These are the channels of E.Reticulum which have attached ribosomes on the surface. It is involved in protein Synthesis.

- **Smooth endoplasmic Reticulum.**

These are the channels of E.Reticulum which donot have attached ribosomes on the surface. These are involved in the metabolism of lipids & detoxification of toxic compounds.

Major functions of E. Reticulum.

⇒ It provide machanical Support.
⇒ R.E.R involved in protein Synthesis.
⇒ S.E.R metabolism of lipids. & detoxification of toxic compounds
⇒ It also involves in transport of material in the cell.
⇒ In Some Cells S.E.R help in the transmission of nerve impulse & Storage of Calcium ions in muscle cells.

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Golgi apparatus. (CSS, PMS)

Dafination.

It is a complex of vessicles & folded membrane's within the Cytoplasm of most eukaryotic cells. Discovered by Camillo Golgi in 1898.

⇒ Cisternae stacks are formed by the fusion of vesicle's while the vesicle's are formed by the budding of S.E.R.

faces of Golgi Complex.

⇒ Outer Convex surface of cisternae stacks in a forming face.

⇒ inner Concave surface is a maturing face. Vasicles are formed in the maturing face.

⇒ The whole stack is consists of a number of cisternae which move from outer to inner face.

Composition.

Chemically Golgi membrane's are similar to plasma membrane. Golgi complex has protein, Corbohydrates Glyco proteins & lipids.

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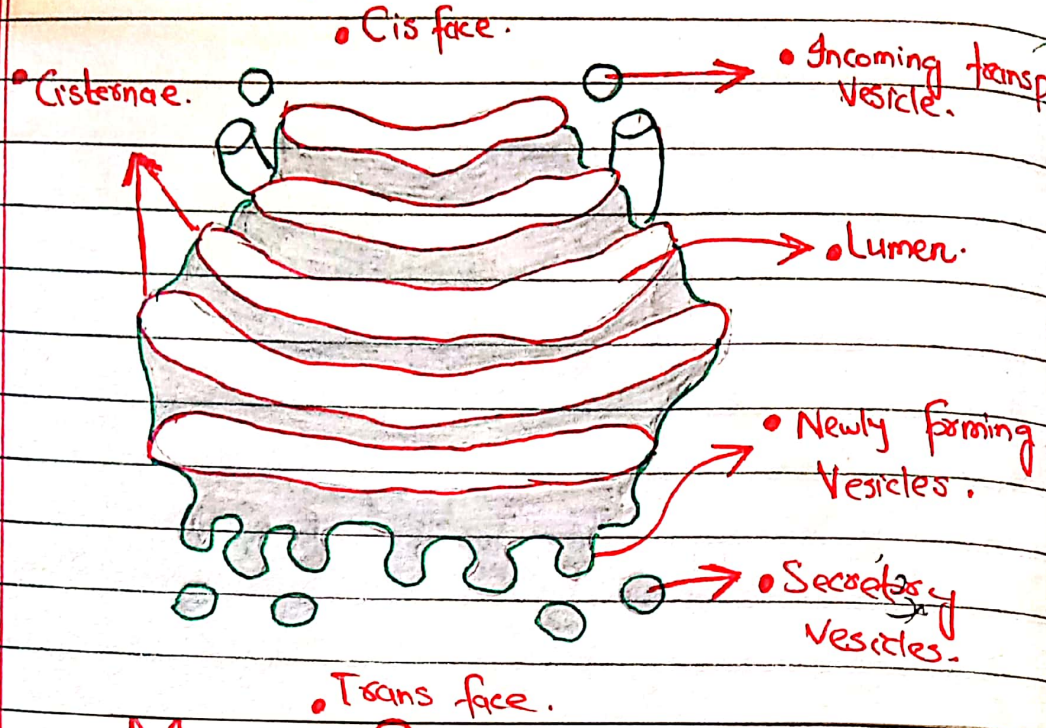

Major function's of Golgi

- formation of Conjugated molecules.

The most important function is the Conversion of proteins & lips to glycoproteins or glycolipid's by adding Carbohydrate's to them.

- Cell Secretion.

Golgi apparatus is also concerned with cell Secretions, eg. Pancrease secreate granule's containg enzyme's that help in digestion. These granules are formed by Golgi apparatus. Proteins & enzymes also passe out the G. Apparatus

- Cell wall Synthesis.

Aslo involved in Cell wall Synthesis

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Lysosomes (CSS, PMS,)

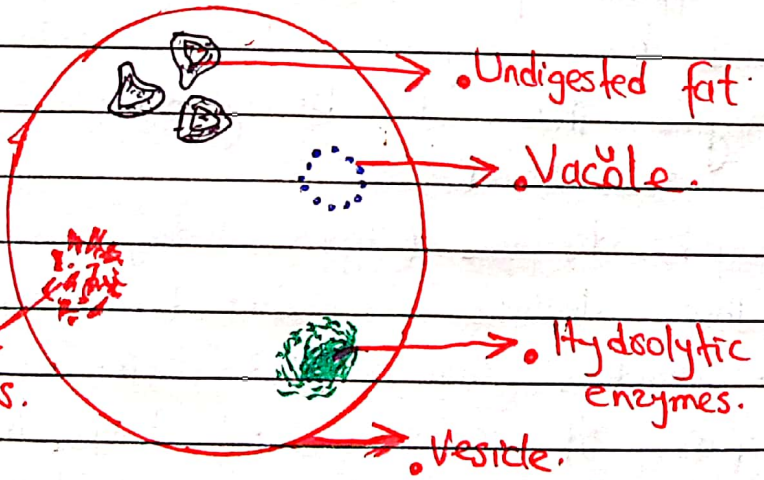
Dafination..

Lysosomes are membrane bound organelles which are spherical in shape. & contain enzymes called acid hydrolyses.. which are capable of digesting organic molecules (lipids, proteins nucliec acids & polysaccharides) under acidic Conditions.

formation

Its enzymes are Synthesized on R.E.R. furthur processed in Golgi Complex & then buffed out as Golgi vessicles, Thus exposing the vesicle's Contents to the lysosomal enzymes which digest them.

=> These are most abundant in the cells, which shows phagocytic activities.



Lysosomes.

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Major function's of the lysosomes.

- lysosome's release enzyme for extra-Cellular digestion.
- lysosomes involve in phagocytosis.
- Also in involved in intracellular digestion
- lysosome's help in Autophagy (it is a process in which Old dead or injured part of the cell are digested)
- lysosome's are involved in degeneration of Cell during developmental process such as metamorphosis.

Diseases caused by lysosomal Abnormalities.

- Storage disease's.

The diseases which are caused by lysosomal abnormalities are called Storage disease's.

- Glycogenosis Type-II (due to absence of D-Glucosidase) In this disease liver & muscle appear filled with glycogen within membrane bound organellies. D-Glucosidase the enzyme that decrades glycogen to glucose is absent.
- Tay-Sach disease. (lipid storage in brain cells) This disease is due to absence of hexa~~m~~seamidase that is involved in Catabolism of lipid's. It causes lipid storage in brain cells which leads to mental retardation & death.

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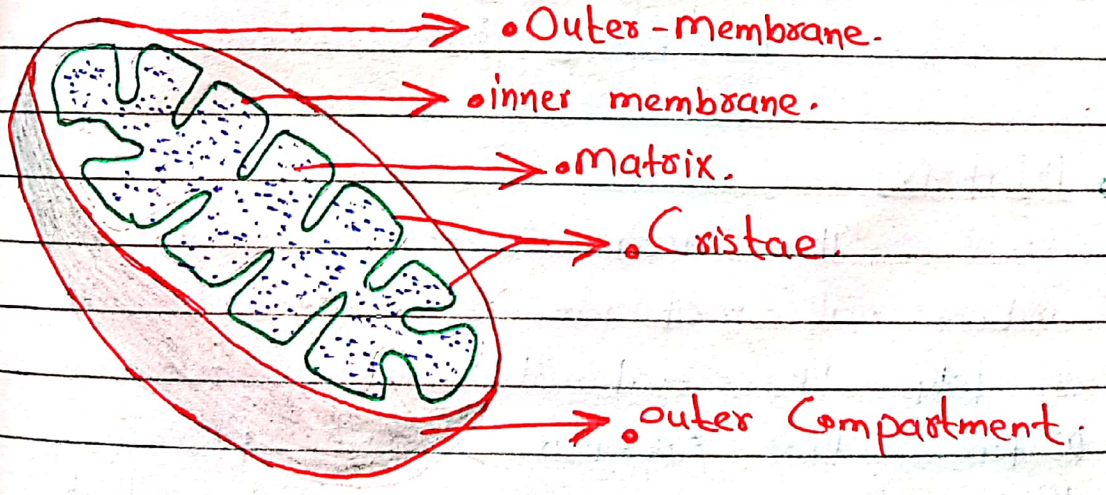
Mitochondria. (CSS, PMS,)

Dafination.

Mitochondria are cytoplasmic double membrane bound organelle present in all eukaryotic cells except R.B.Cells. also known as power house of Cell.

Structure & Size.

Mitochondria varies in size & shape both. In generally they are filamentous or granular double membrane organellies. In term's of size mitochondria also varies. e.g. pancreatic cells contain mitochondria of 10μm & Ocyte's in Amphibian Contain's 20 to 40μm long mitochondria but normaly it range's in 2μm to 8μm.



Mitochondria.

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• Numbers..

Cells which utilize high amount of energy have high number of mitochondria. Most Vertebrates have 500 to 1000 mitochondria in a cell. In human's number of mitochondria varies 2 to 2,500 in cell.. depending on tissue type or other factors. Oocytes of Amphibians Contain 300000 m...

Structure of Mitochondria..

• Membranes...

Mitochondria is Sorrounded by double membranes. Outer membrane is about 60 A° thick & work as boundary between Cytoplasm & mitochondrial matrix. The inner membrane is also 60 A°. & is folded like finger like structures Called Cristae. Cristae increases Surface area for multienzyme system. It also Creat Space b/w inner & outer membrane is called Intermembrane space.

• Matrix

The area covered by inner mitochondrial membrane is filled with a jelly like material called matrix. Matrix Contains ribosomes & DNA. Presence of these things on matrix indicates that mitochondria can synthesize its own protein. That why mitochondria also called Semi Autonomous body

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Sub-Mitochondrial particles.

There are many Lollipop like structure's are present in mitochondria called Sub-mitochondrial particle's. These particles Consist of 3 parts.

- **Head or Coupling factor (F1).** It is composed of 10 proteins in 4 different forms. (types). Which act as enzyme in phosphorylation of ADP to ATP.
- **Stalk or factor "O".** factor O connect F1 to membrane. This factor is composed of protein's why regulate ATP Synthesis.
- **Membrane piece or base.** These membrane piece function's as proton channel through which proton flows back from intermembranes space to matrix. These proton's generate's proton motive force (PMF) which is used in phosphorylation of ADP→ATP.

Do you know?

Due to presence of ribosome's & DNA on mitochondria. It is self replicating organelle. (New mitochondria are formed by the division of Old mitochondria).

Major function's of mitochondria.

- Involves in protein Synthesis.
- It provide site for oxidation-reduction reaction, of food to produce energy.

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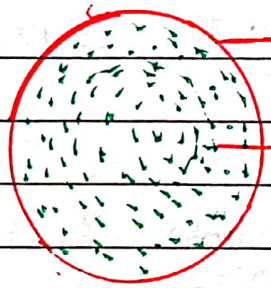
→ Mitochondria also called power house of cell because it is concerned with Synthesis & supplies of energy for various function's of Cell.

Vacuole (CSS, PMS)

- **Dafination.**

Vacuole's are sacs which are sorrounded by memembranes and are part of Cytomembrane System. Vacuole's occure in different shapes and size's.



Vacuole

- **Major functions:**

- **Storage of food.**

Some protozoa & sponge's have vacuole for storage of food.

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- Homeostasis.

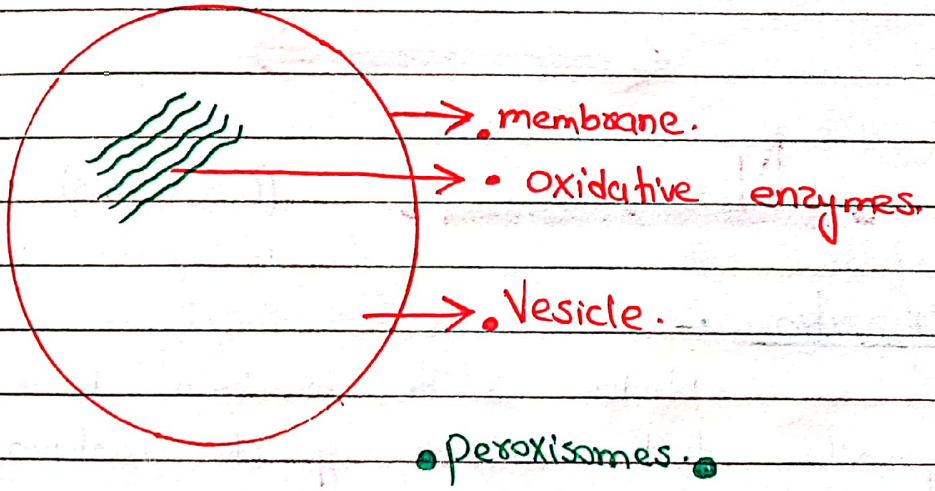
In animal like protozoa & sponge's, Contractile vacole are present that Contract water from Cytoplasm & pump it to the outside to maintain its internal enveronment.

- In plants a large water vacuole maintains turgidity.

Peroxisomes. (CSS, PMS)

- Dafination.

These are enzyme's containing vesicle's present in the Cytoplasm.



- Composition.

They Contain different enzyme's peroxidase, oxidase, catalase & glyolic acid oxidase e.t.c.

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- **Presence.**

Peroxisome's are present in all type's of Cells, protozoan's, Yeast & many higher plants but are abundantly present in liver Cells.
They are derived from E. Retraulum.

Major functions.

=> Animal's proxisome's participate in number of process's involving oxidation.
The enzyme Catalase catalyzed that decomposition of Hydrogen peroxide's to water & oxygen.
=> In plants they are site for Glyoxylate Cycle.

Nucleus. (CSS, PMS)

- **Defination.:-**

Nucleus is a membrane-bound organelle that contain's genetic material (DNA) of eukaryotic organisms.
Nucleus is discovered by Robert brown in 1831.
Nucleus can be best studied in non-dividing Cell's because in dividing cells nucleus Chromatin material change's into Chromosomes.

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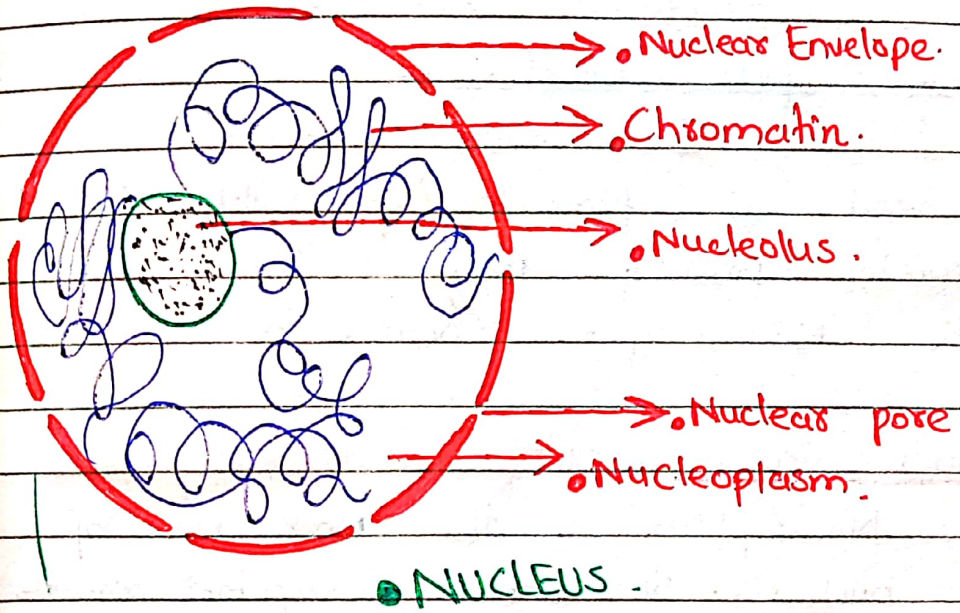
Structure of Nucleus. :-

Nucleus Consist of the following structures.

- Nuclear envelope
- nucleoplasm.



- NUCLEUS.

Nuclear envelope. (1)

It is a membrane that separates the nuclear material from the Cytoplasm It is Composed of double membrane.

- Outer membrane → It links with E.Reticulum.
- Inner membrane → It encloses nuclear material.

Nuclear pore's.

The nuclear envelope is porous its number of pores varies in different cells.

- In Undifferentiated Cells. (growing cells & egg cells). have numerous pores (3000 per nucleus.)
- In differentiated Cells. (Red blood cells) have less pores 3-4 pores per nucleus.

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- **functions of Nuclear pores.**

It Control the transport of molecules into & out of the Cell's nucleus. The Size of pore's is very small so, it prevents leaving of DNA from nucleus.

- **Nuclear pore Complex. :-**

Each nuclear pore is composed of filamentous or Globular granules "probably protein's" These granules from Nuclear pore Complex.

- **Nucleoplasm. :- (ii).**

Nucleoplasm is the inner mass of the nucleus. It Contain's 2 important organelles.

- Nucleolus.
- Chromosome.

- **Nucleolus. :-**

It is a small dense spherical structure in the nucleus.

- **Structure. :-**

It is Composed of 2 regions.

- The peripheral region.

The peripheral region containing precursors of ribosomal Subunits.

- The Central fibrillar region.

The Central fibrillar region contains large molecular weight DNA or RNA.

- **functions.**

It Contol & activates ... or

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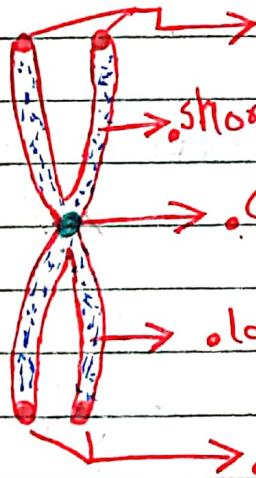
regulates the various activities of Cell. (eg. Growth & metabolism) & Contain hereditary information.

- Do You know?
  - The nucleus was the first of cell organelles to be discovered by the Scientist. in (1831).
  - It contains 10% voleum of the cell.

- **Chromosomes.** :-

Chromosomes are thread like structures located inside the nucleus of animal or plant Cell. (OR.) During the Cell division the Chromatin material is converted into thread like structures called Chromosomes.



- No. of Chromosome
  - Human → 46.
  - frog → 26
  - Drosophila → 8
  - Snail → 24
  - Tiger → 38
  - Lion → 38

- **Structure.**
- **Chromatids.** :- Each Chromosome consist of 2 Chromatics which are attached to each other at a point Called Centromere. During Cell division Spindle fibers are attached at Centromere point.
- **Information points.** :- Informations about Cellular activities are located on the

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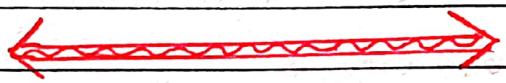
Chromosome's in the form of genes.

- **Role of Chromosomes :-**
  Help in transfer of genetic material or information from One generation to Next.

Major functions of Nucleus.

⇒ Nucleus store's genetic information's. These informations are transfered to the next generation during Cell division.

⇒ The nucleus also directs Chemical reactions in the Cell by transcribing Genetic information from DNA to RNA.



Cytoskeleton. (CSS, PMS)

- **Defination :-**
  Cytoskeleton is the network of filament's & tubule's that extends throughout the Cell, through the Cytoplasm. Connecting the various organelle's & Cellular components.

- **Chemical Composition. :-**
  Chemically the cytoskeleton is composed of Contractile protein's like, Tubulin, Actin, Troponin & tropomyosin.

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Structure of Cytoskeleton.

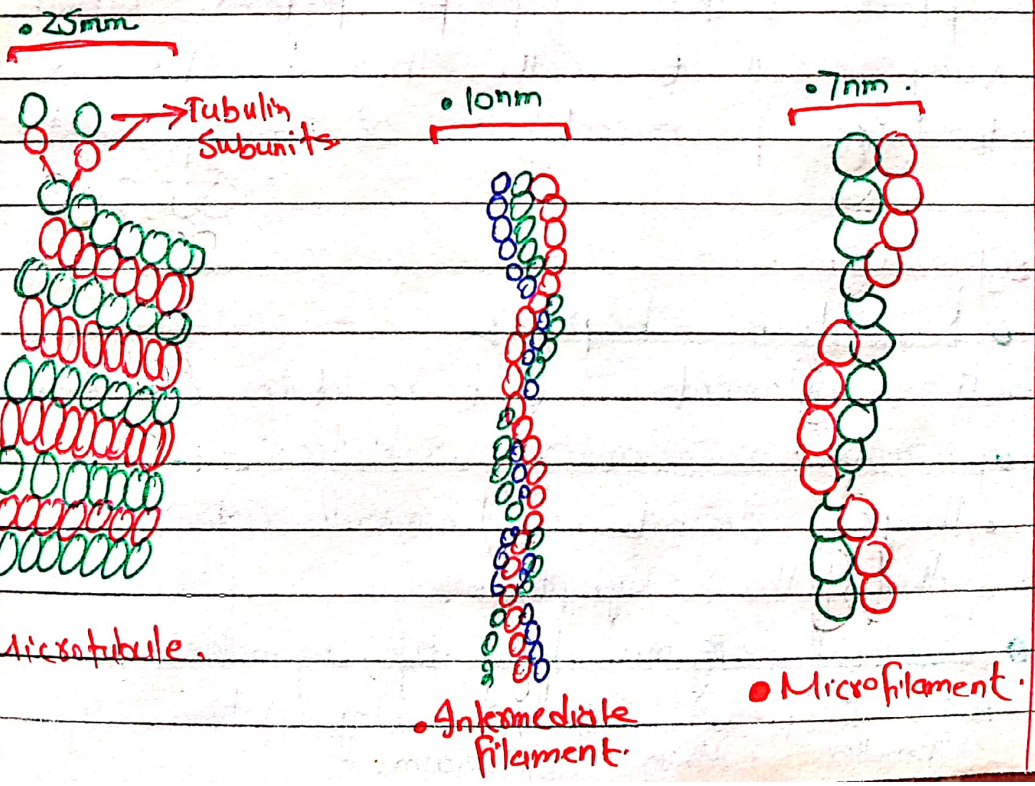
- The Cytoskeleton structure's are:-
- Microtubules.
- Microfilaments.
- Intermediate filaments.

Microtubule's

These are hollow, long, unbranched, slender & fine cylindrical structures in animal. which are made up of Tubulin Sub-units.

- functions-

- Movement of organelle's. Microtubule's help in the movement of organelles - eg Secretory Vesicles.
- part of transport System. They are also a part of transport system.



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Microfilaments :-

Microfilaments are solid strings of actin protein molecules. which are mostly seen in muscle cells & help in muscle contraction.

functions. :-

- Mechanical Support. In non-muscle cells actin microfilaments provide mechanical support for various cellular structures.
- form Contractile Systems. They also help to form contractile systems responsible for some cellular movements. (eg. Amoeboid movements)

Intermediate filaments :-

These are chemically heterogenous group of protein fibers, the specific proteins which vary cell to cell. Their diameter is in b/w of microtubules & microfilaments.

functions.

- Maintain shape. :-
  - These filaments help to maintain cell shape.
- promote mechanical activities. :-
  - They promote mechanical activities within the Cytoplasm.
- Spacial organization of organelles. :-

Also help in spatial organization of organelles within Cytoplasm.

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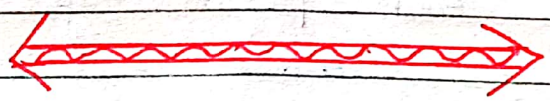
Overall Cytoskeleton functions.

- They are involved in endocytosis & exocytosis.
- Help in Amoeboid movement of Cell.
- Actin & myosin filaments cause the muscle Contraction
- Microfilaments involved in internal cell movements.
- The formation & removal of spindle fibers is due to micro-tubules.



Plasma Membrane. (CSS, PMS)

Defination.

Outermost layer in the animal cell which is delicate, porous, thin & elastic. is known as plasma membrane.

Chemical Composition :-

Chemically, it is composed of proteins & lipids. The proteins are 60 – 80% & lipids are 20 – 40%. Chorbohydrates & Colestrol are also present in small amount.

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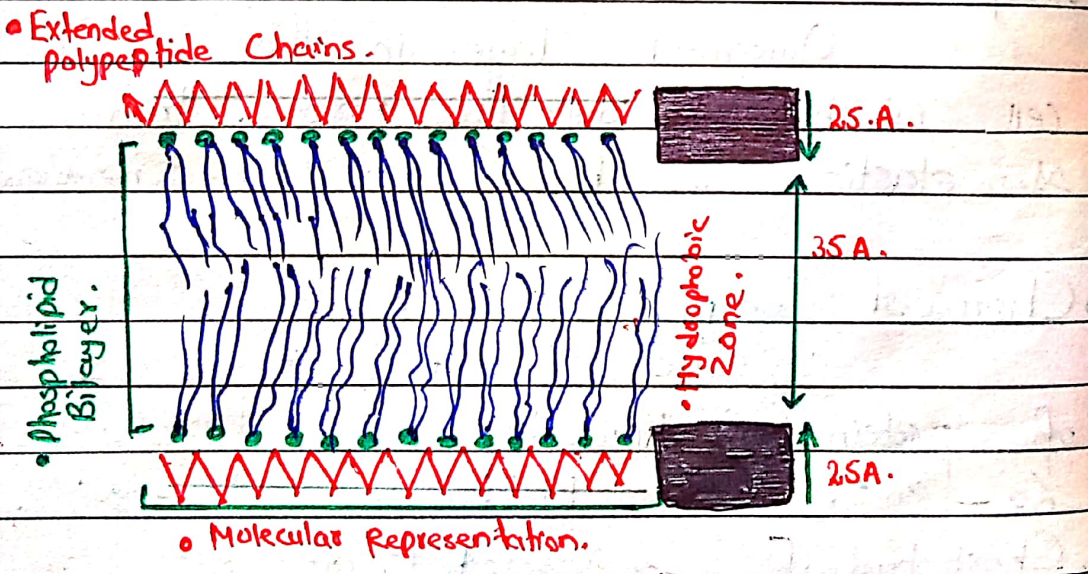
• Structure.

In plasma membrane the protein & lipid molecules are arranged in specific arrangement.

• Unit Membrane Model.

This model was given by Danielli & Dawson in 1935. Firstly, it was proposed that the cell membrane is composed of lipid bilayer sandwiched b/w 2 protein layers. This model is known as. lipid bilayer sandwich model.

- Thickness of each protein layer is 20 – 25 A & middle layer is 35 A thick. This basic structure is called unit membrane & is found in all cellular organelles.



• Unit Membrane Model.

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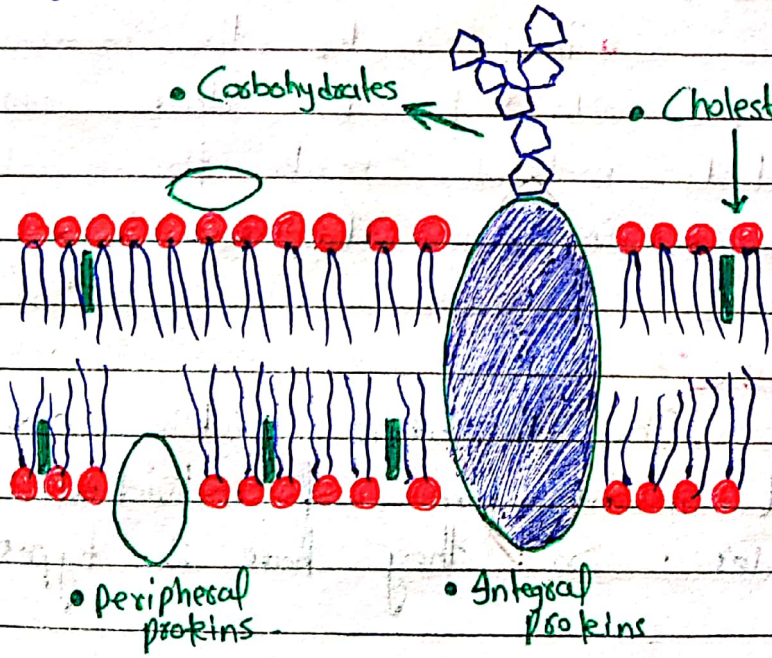
• NOTED.•

The modern technology has revealed that lipid bilayer is not Sandwiched b/w 2 protein layers.

• Fluid Mosaic Model.•

This model was given by S. jonathan, Singer & Garth Nicolson. According to this model in structure of membrane the protein are embedded in the bilayer of phospholipids in a masaic manner & is fluid rather than Solid. On the basis of the Continuosly changing position of molecules but maintaining the sheet like Structure, named as the fluid Mosaic Model The word mosaic refers to the many different kinds of protein's dispersed in the phospholipids bilayer.



• Fluid Mosaic Model.•

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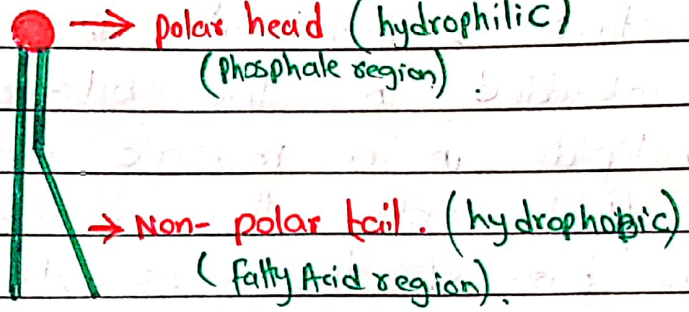
(i) Phospholipids :-

The phospholipids have 2 ends.

- polar end or head :- It has phosphate group. They are called (Hydrophilic) (Water attracting)
- Non-Polar end or tail :- It has fatty acids group or molecules. They are called (Hydrophobic) (Water repelling).



(ii) Cholestrol :-

Cholestrol molecules are embedded in the interior of plasma membrane.

- functions.
  - They make the membrane less permeable for water soluble Substances.
- Cholestrol molecules make the membrane rigid & increase its stability.

(iii) Proteins :-

The membrane proteins are present in the form of individual molecules. So they have 2 types.

- Peripheral proteins :- These membrane are

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attached to inner & outer membrane surface.

- Intrinsic proteins :- These proteins are embedded in lipid's bilayer.

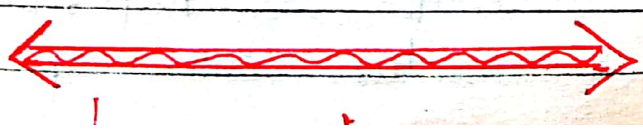
- functions.
  - Some intrinsic proteins move ion's or molecule's across the membrane.
  - Some proteins attach the membrane to the cell's inner Cytoskeleton.

(iv) Carbohydrates :-

In plasma membrane Carbohydrates are present in 2 forms.

- Glycoproteins:- The Carbohydrate's unite with protein's & form glycoproteins.
- Glyco-lipids. The Carbohydrate's unite with lipid's & form glycolipids.
- Glycocalyx (cell coat). Surface Carbohydrate's & portion's of the protein's & lipid's make up the glycocalyx or "Cell Coat"

- functions :-
  - Glycocalyx is necessary for cell recognition.
  - Some Carbohydrates behave as key Component in Co-ordinating cell behaviour in animals.



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Date: __/__/20

(30)

Nervous Tissues. (only CSS)

- **Defination.**

Tissues which are specialized for reception of stimulie & Conduction of impulses from one part or region of the body to another body.

- **Types.**

Tissues are not included in PMS.

- **Neurons.** (i)

It is basic functional unit of nervous system. Neurons in nervous tissue transmits nerve impulses or electrical signals to the other neurons, muscles or glands.

- **Neuroglia** (ii)

A variety of non-nervous cells that insulate neuron membranes & Serve different supportive functions.

- **Peripheral glial Cell.** (iii)

The cells that form sheaths & help, protect nourish & maintain cells of

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peripheral nervous system are called peripheral glial cells.

Nervous Tissue.

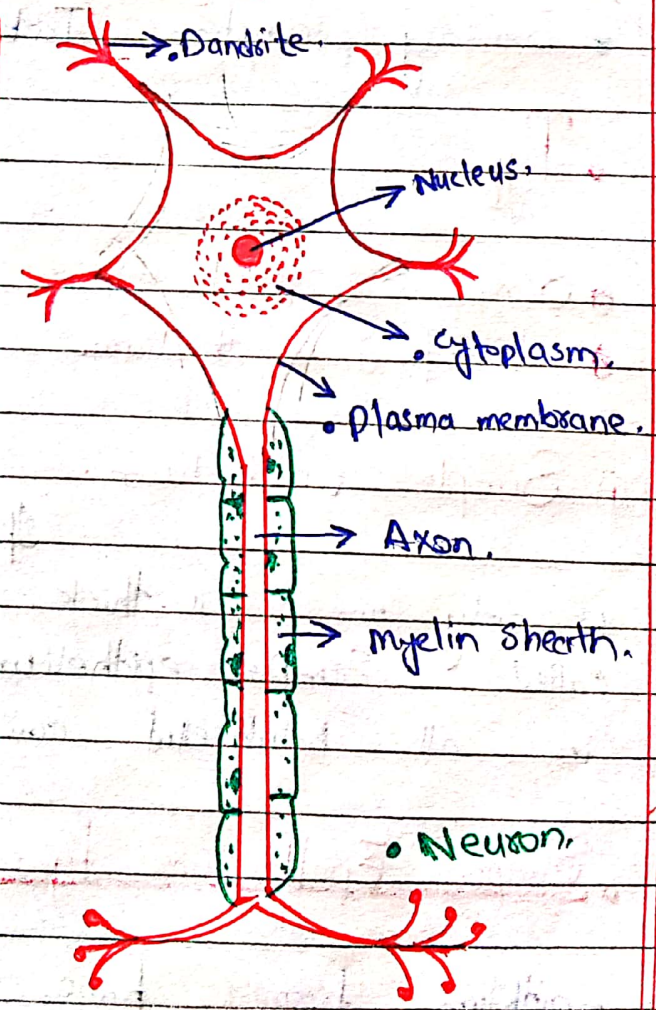
- Neurons
- Neuroglia Cells.

- Cell body.
- Nerve fibers.

- Medullated.
- Non-Medulated.

Do You know?

- Neurons are the oldest & longest cells in the body.
- The longest axon of a human motor neuron can be over a meter long. reaching from the base of spine to the toes. length upto 1.5 meter.



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(Only CSS)

Epithelial Tissue & Types.

- Dafination.

These are tissue's which cover's the outside of body & lines organ's & Cavities. The Cells in this kind of tissue are very closed packed together.

Major functions.

- Absorption of material. (Small intestine lining)
- Transport. (Kidney tubules)
- protection. (Sweat glands)
- Ecretion. (Skin)
- Sensory perception. (Taste buds in tongue)

Types.

- Simple epithelium.
- Stratified epithelium.

(i) Simple Epithelium.

If the epithelium is only one-Cell thick the epithilium is called Simple epithelium. It is found in all Multicellul animals except Sponges.

Simple Epithelium types.

Simple epithilum tissue's have 4 types.

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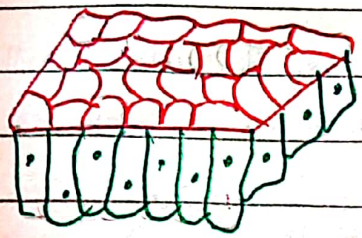
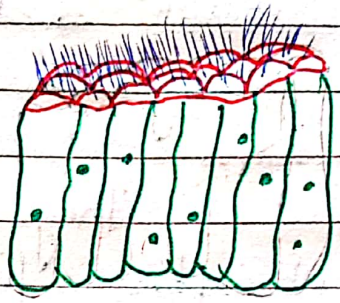
• Cuboidal epithelium.

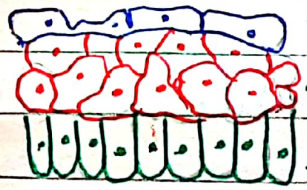
• Squamous epithelium.



• Simple Columnar epithelium.



• pseudo-stratified Ciliated Columnar epithelium.



• Transitional epithelium.

• Simple Squamous epithelium.

The Cells of Squamous epithilium are extremely thin & flat. All the cells are arranged edge to edge. It forms delicate lining or Covering

• Location.

Alveoli in lungs, Bowmens Capsule in kidney, inner year of lining of blood vessels.

• function.

These tissues are protective in functions.

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(ii) Simple Cuboidal epithelium.

- Structure.

It is composed of short box like cells.

- Location.

Cuboidal epithelium usally lines small ducts & tubes such as in kidney & Slivery glands.

- function.

They have active secretory or absorptive functions.

(iii) Simple Columnar Epithelium.

- Structure.

The cells of Columnar epithilium are elongated

- location.

forms in the mucous membrane of the stomach, intestine & urinogenital organs & their ducts.

- function.

Absorption & Secretion.

- Simple pseudo- Stratified epithelium.
- Structure.

It is a type of epithelium that though Comprising only a single layer of cells,

- location.

these are found in mucous membrane of olfactory chambers & in

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the trachea.

- function.

Propels mucous or reproductive cells by Ciliary action.

- (Types) Stratified epithelium.

The epithelium Consisting of more staked layers is Called Stratified epithelium.

- Stratified squamous's epithelium.
- Structure.

These cells consist of Squamous epithelial (flattened) cells arranged in layers upon a basal membrane.

- location.

This type of epithelium line's the oral cavity, esophagus & anal canal of many vertebrate's & Vegina of mammal's.

- function.

It Consist's of 2 to many layer's of Cells.

- Do You Know?
  - Most Common type of epithelium Cells which are found in our body is Stratified squamous epithelium.

- Do You know?
  - Squamous epithelium is the thinest epithelial cell.
  - Karatinized Stratified epithelium are thickest epitheliumcells.

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- Transitional epithelium.
- Structure.

This type of epithelium tissue consist of multiple layers of epithelial cells. which can Contract & expand in distension needed.

- Location.

This epithilium is found in urinary tract & bladder of Vertebrates.

- function.

Its important function is that it allows organ to stretch & increase its voleum depending on fluid pressure.

written by G.H. Babar
MSc. Zoology. UAF
0308-1127642.

Muscle Tissue. (only CSS)

- Dafination.

Muscle Tissues consist of fibers of muscle cell's connected together in sheet's & fibers.

- Involuntary Muscle's.

The Contraction of involuntary muscle's is not

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under the Control of our will.
e.g. Smooth & Cardiac muscle.

- Voluntary Muscle's.

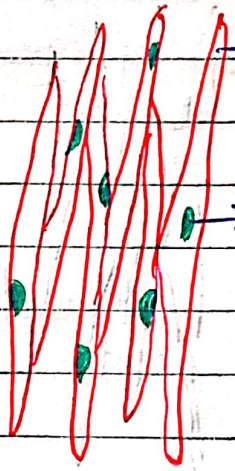
The contraction of voluntary muscle's is under the Control of our will. e.g. Skeletal muscle's.

Note

(Exercise does not increase the number of our skeleton muscle cells it simple elarge those already exists.)

- Type's

- Skeletal muscle.



- Smooth muscle.

- Cardiac muscle.

Do You Know?

- Skeletal muscle fiber tissue is the largest tissue in human body.
- Cardiac muscle's are hardest working muscle's in human body.

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Skeletal Muscles

- **Structure:-**
  Composed of striated cells that are long & cylindrical & contain many nuclie.
- **Location:-**
  found attached to bones e.g. bicep muscle's.
- **function:-**
  Responsible for voluntary actions & locomotion.

Smooth Muscles.

- **Structure:-**
  Smooth tissue are composed of non-striated (smooth) cells that are spindle shaped & each contain a single nucleus.
- **location:-**
  found in the walls of digestive tract, Urinary bladder & blood vessel's.
- **function:-**
  Help in movement of substance's (food stuff, urine ...) along internal passageways.

Cardiac Muscles.

- **Structure:-**
  Composed of striated cells that are that are branched &

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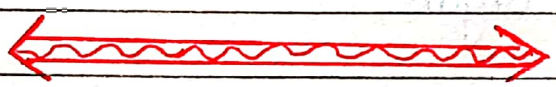
each contain a single nucleus.

- Location :-

found in the walls of heart.

- function :-

produce's heartbeat that popel's blood into the Circulation.



Connective Tissues. (only CSS)

- Dafination.

Connective tissue's are diverse group of tissue's that Serve vasious binding & Supportive function's
Connective tissue is relatively composed of relatively few cells, a great many extracellular fiber's & ground Substance's also called matrix in which the fiber's are embedded.

- Type's.

- Loose Connective tissue.

It is composed of fiber's, Both fixed & wandering cells Suspended in a viscous fluid ground Substance's.

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Functions.

They bind Epithelia (Continuous sheets of Cells that Covers the exterior Surface of body, line internal Closed Cavities of body tubes that Communicate with outside environment e.g. alimentary & respiratory canals) to underlying tissues.
They are used as packing material to hold organ.

fibers connective tissues.

The Collagen fibers are densely packed in fibrous connective tissue. These fibers lie parallel to each other. So they Creat very strong Cords. These are.....

Tendons.

- Tendons have pre-dominance of Collagen fibers
- fibrocytes are fund.
- Tough & inelastic.
- join the skeleton muscle to bones.
- fibers are regularly arranged.

Ligaments.

- Ligaments have pre-dominance of elastic fibers.
- fibroblasts are found.
- Strong but elastic.
- Connect bones to bones at the joints.
- fibers are irregularly arranged.

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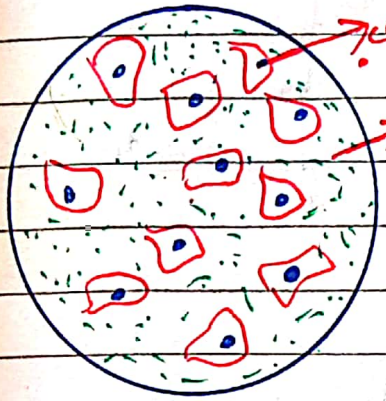
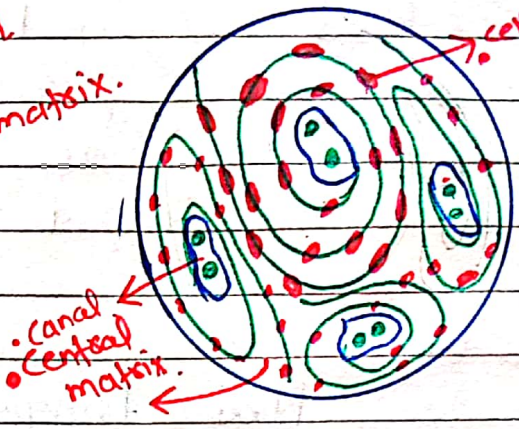
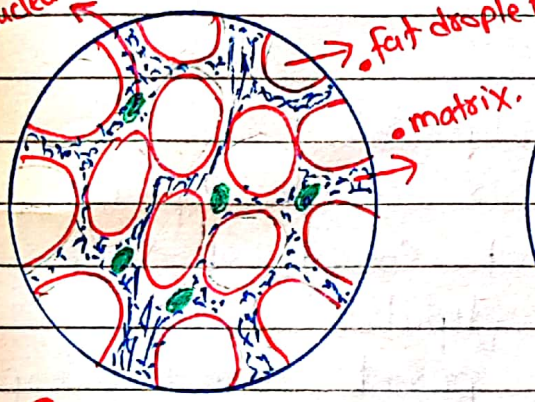
- Cartilage.



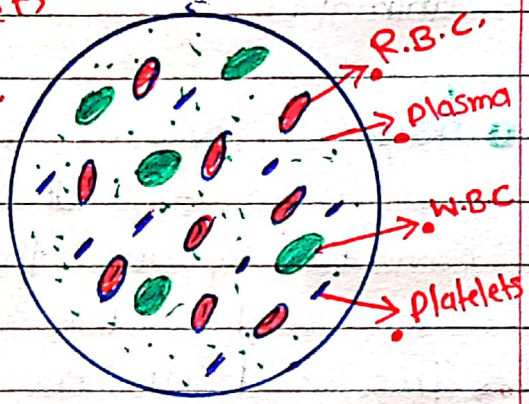
- Bone.



- Adipose tissue



- Blood



Cartilage.	Bone.
Soft, elastic & flexible Connective tissue.	Tough, rigid & inelastic Connective tissue.
Cartilage cells are Called (Chondrocytes)	Bone cells are Called. (Osteocytes)
Matrix doe's not Contain the deposition of Calcium Salts.	Matrix alway Contain deposition of Calcium Salts.
Matrix lack blood Vessel's	Matrix have blood Vessels.
Bone marrow is absent	Bone marrow is present.
3 types.	
Hylin, elastic & fibro Cartilage.	
Found in nose & external ear	

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Blood.

Blood is connective tissue. It has fluid called plasma. In plasma red & whit blood cells along with plaletels are present.

function:

Blood transports various Substances throughout the bodies of animals.

Adipose tissue. (Adipocytes)

It is a type of loose connective tissue that consist of large cells and store lipids.

⇒ These tissues are present around kidneys, under the skin, in bones, within abdomen & in breast.

function.

- These provide reserve fuel (Lipids).
- Insulate against heat loss
- provide Support & protection to organs.

Do you know

Mammalian R.B.C contains neither nucleus nor mitochondria.

Do you know?

Bone & Cartilage are most abundent type of Connective tissue in humanbody.

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(CSS, PMS)

Organ & Organ System.

What is Organ.

An organ is a self-contained group of tissue's that perform's a specific function in the body.
e.g. heart, kidney, Liver & Stomach.

> **Do You Know ?**
> The word Organ derived from Latin word "Organum". which mean's "Instrument."

Organ System.

Organ System is an association of organ's that together perform an overall function's.
Many organ's which have different function's Combine to form Organ System.

Type's or Organ System.

- Muscular System.
  - functions.
    - Maintainance of posture, (Structure)
    - Movement of body & body parts.
- Digestive System.
  - function.
    - Capture and physical / Chemical digestion of food.

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- Respiratory System.
- function.
  - Exchange of gases.
  - production of Sound.

- Reproductive System.
- function.
  - perpetuation of Species.

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- Excretory System.
- function.
  - Maintaining proper water balance.
  - Excreating toxic metabolic products as waste.

- Endocrine System.
- function.
  - Regulation of body activities through hormone's.

- Circulatory System.
- function.
  - Transport of materials to cell. & from Cell e.g Co_2 & Urea.

- Nervous System.
- function.
  - Regulation of body activities through Conduction of wave of protoplasmic change in individual neuron's.

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- Integumentary System.
- function.
  - prevent underlying tissue from mechanical injury & dasication.
  - Help in secreation of specific metabolites along with absorbtion.

Do You know?
Skin is the largest organ in our body.

Do you know?
Pineal gland (present in ventral part of brain is the smallest organ in human body)

Explain Mitosis. (CSS, PMS)

- Dafinaiton
  It is a type of Cell division which ensures the same number of Chromosomes in daughter Cells as that in the parent cell.

- Events of Mitosis.
  - Mitosis is mainly devided into 2 Stages.
  - Karyokinesis :- Division of nucleus.
  - Cytokinesis :- Division of Cytoplasm.

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(Karyokinesis.)

- **Mitotic Apparatus.**

The Specialized microtubule structure including aster & Spindle is Called mitotic apparatus.

- first Step of Karyokinesis is the formation of mitotic apparatus.

- **Steps in mitotic apparatus formation.**

(i) The Centrioles are duplicated during interphase. But both Centrioles are present in the same centrosome. The position of these Centrioles takes place at the bigginig of Karyokinesis. Two pair of Centrioles separate & migrate to the opposite sides of the nucleus. They establish bipolarity of the establishing Cell's.

(ii). Sets of microtubules (fiber) originate from each pair of Centrioles. These microtubules are composed of a protein tubulin & traces of RNA. This specialized microtubule structure including aster and spindle microtubule. is called Mitotic Apparatus.

- **Function of Mitotic Apparatus.**

It is used to attach & Capture Chromosomes. So, they align & Separate the Chromosomes in equal distribution way.

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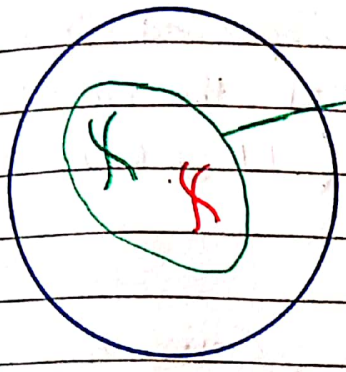
Day: M T W T F S

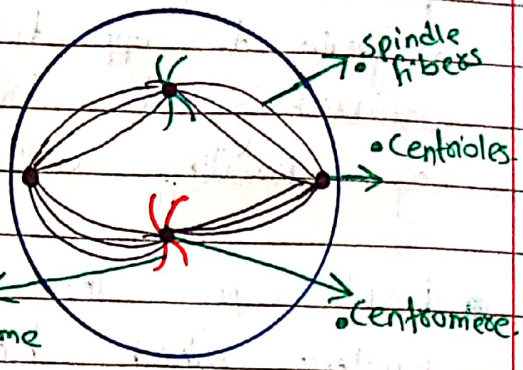
Karyokinesis (4n)

Date: ___/___/20___

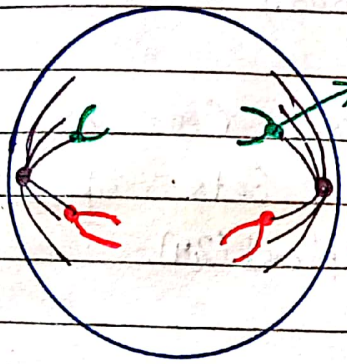
- Prophase.



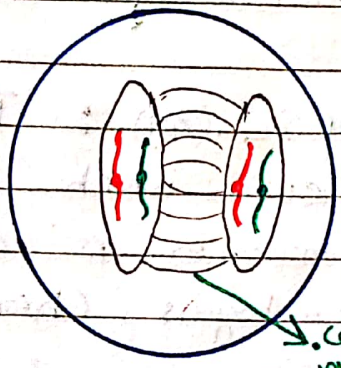
- Metaphase.



- Anaphase.



- Telophase.



• Some basic term's used in Meiosis & Mitosis.

- **Centriole** :- One pair of cellular organelles that occure especialy in animals are found near the nucleu's function in formation of spindle apparatus during cell division.

- **Centrosome** :- Centrosomes are organelle's that serve as the main micro-tubule organising ceinter's for animal cell. & it helps organize the spindle fibers during mitosis.

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- **Bipolarity :-** Early in mitosis 2 pairs of Centrioles Separated. They migrate to opposite sides of the nucleus. establishing the bipolarity (دو قطبیت) of dividing Cell.

- **Haploid (N).** The organism, Cells or nucleus which has a single set of Chromosome's. is called Haploid. (N) (Non-Homologous Chromosomes.)

- **Diploid (2N).:** The organism Cell's or a nucleus which has 2 Sets of Chromosome's is called diploid (2N). (Homologous Chromosomes)

- **Do You Know?**

  In Our body About 2-5 million Cells undergo mitotic division every Second. (2500000)

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Sub Stage's of Karyokinesis.

→ Karyokinesis is futhur divided into
- Prophase.
- Metaphase
- Anaphase.
- Telophase.

Prophase. (pro - 1st phage - stage = first stage)

The chromosomes are not visible even with electron microscope during Interphase (Non-dividing) of cell cycle. However by Using histological strain's for DNA. A network of very fine thread's can be visualized. This network is Called Chromatin.

→ following Change's takes place in Chromatin during prophage.
- The Chromatin material is Condensed by fiding. Thus Chromosomes appear as thin thread's. (0.25 μm — 50 μm in length) at the biggining of prophage.
- Chromosome's become thicker by fiding. Ultimately Chromosome's become visible. having 2 Sister Chromatid's attached at Centromere.
- The nuclear envelop disappear's at the end of prophase & nuclear material is released into Cytoplasm.

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- Nucleoli disappears.
- Mitotic apparatus Organized
- Cytoplasm become more Viscous.

Metaphase :- (Meta, middle phase. stage.)

- Now, each metaphase Chromosome has duplicated structure. It Consist of 2 Sister Chromatids. The Chromatids are attached at point called Centromere. or primary Constriction.
- The Centromere has special Area Called Kinatochore. Kinatochore has specific base arrangements of Special protein's where Kinetochore fibers of mitotic apparatus attach.
- The Kinatochore fiber of spindle attach at the Kinatochore of Chromosome. the fibers arrange the Chromsomes at the equator of the spindle thus they fom equatorial plate or metaphase plate in the equator.
- Each Kinetochore is attached with 2 fiber's each from opposite pole's

Anaphase. :- (upward movement phase)

- It is the most Critical phase of mitosis. It ensure's the equal distribution of Chromatid's in daughter Cells.
- The Kinatochore fibers of Spindle

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Contract towards their respective poles. by exerting force on Centromere & Sister Chromatids are separated from Centromere.

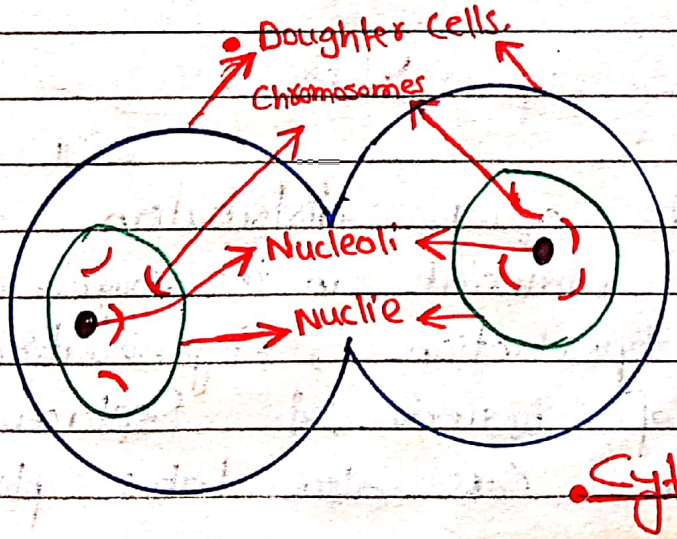
- As a result half of Sister Chromatids travel towards each pole.
- At the same time polar microtubules elongates.

• Telophase. (Telo- Tail → Means Last phase)

- When Chromosomes reach at opposite poles the anaphase ends & telophase begins.
- The Chromosomes Unfold & Uncondensed.
- Mitotic Apparatus disorganized.
- The nuclear membrane & nucleoli are reorganized.
- 2 nuclie are formed at 2 poles of Cell.

• Cytokinesis.

The Division of Whole Cell (mainly Cytoplasm) is called Cytokinesis.



Cytokinesis.

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Cytokinesis in Animal Cells:-

During the late telophase Astral microtubule Send Signal's to equatorial region of Cell. These Signal's activate's the actin & myosin. The actin & myosin form Contractile ring. Their Contraction form's Cleavage furrow. The furrow deepens towards the Center of the Cell's. It finally devide's the parent Cell into 2 daughter Cell's.

Cytokinesis in Plant Cells:-

In plant cells, Golgi vesicle's are Collected at the former metaphase. Membrane Structure Phragmoplast is formed. A cell plate formed by the fusion of the vesicle's of the phragmoplast grows from the Center toward the cell walls & the membranes of the vesicle's fuse to form a plasma membrane that divides the Cell in 2.

Importance of Mitosis.

- It ensures the equal distribution of haiditary material in the daughter cell.
- Asexual reproduction, Regeneration healing of wound's & Replacement of older Cell's also takes place

by mitosis.

- Development and growth of multicellular organisms depend upon orderly Controlled Mitosis.
- Cancer is also caused by uncontrolled Mitosis.

Do You Know?

There are only 4 parts of human body where mitosis can occure Your Skin, Muscles, DNA & Circulatory System.

Written by. CH. Babar.
Msc-Zoology. UAF.
0308-1127642

Explain Meiosis. (CSS, PMS)

Defination.

Special type of Cell division in which the numbers of Chromosome's in daughter cells are reduced to one half as compared to the parent Cell's is Called Meiosis.

- **Occurance.**

It takes place only in diploid Cell's. Meiosis takes place in animals at the time of Gamete

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formation. In plants it take's place during Spore formation. 2 Consecutive division takes place after single replication of DNA during meiosis. So each diploid cell's produce's 4 haploid Cells as a result of meiosis.

There are 2 kind's of Meiosis.

- Meiosis I :- Reduction division.
- Meiosis II :- Same like Mitosis.

Meiosis. I.

Meosis I is furthur divided into the following phases.

- prophase I
- Metaphase I
- Anaphase I
- Telophase I

Prophase.

This is very long phase. The Chromosome behave has homologous pair during meiosis. So, it is totally different from prophase of mitosis. in which Chromosome's are not arranged in homologous pairs ~~during~~ This phase consist of following Stage's.

- Leptotene. (Riban like.)
- Zygotene. (Pairing.)
- Pachytene. (Packing)
- Diplotene. (Repelling)
- Diakenesis. (Condensation)

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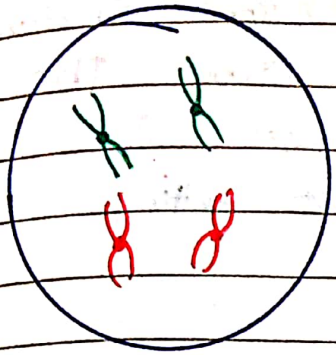
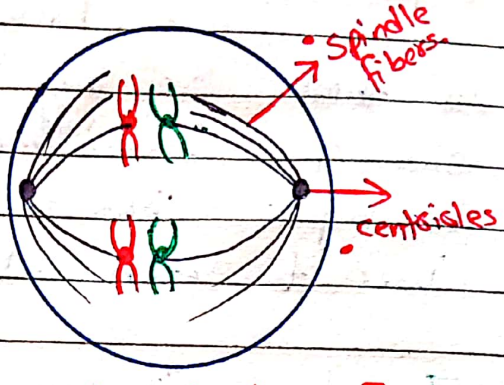
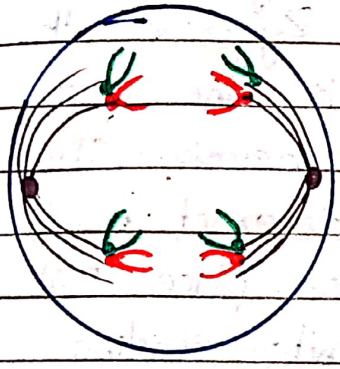
• Meiosis I •



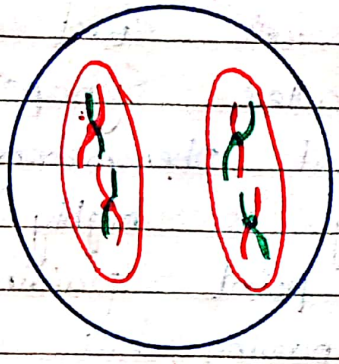
• Prophase I



• Metaphase I

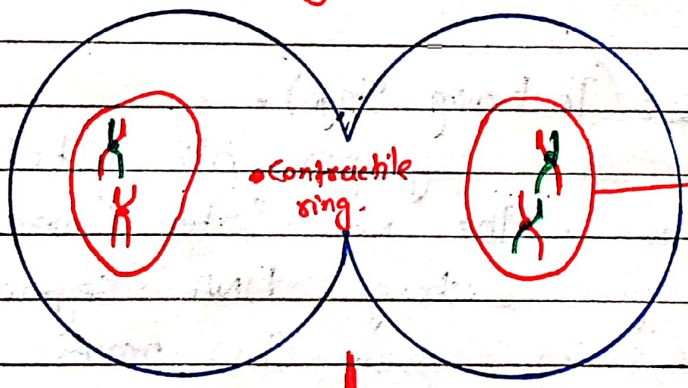


• Anaphase I



• Telophase I

• Cytokinesis :-



Meiosis II

Meiosis II starts after Cytokinesis of Meiosis I which follows the same steps as Mitosis.

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• Leptotene :- (Ribbon like stage).

The Chromosomes become Shorten & thick. So, they become visible. The size of nucleus Chromosome's increase's & homologous Chromosomes start getting closer to each other.

• Zygotene :- (pairing stage).

The pairing of Chromosome's start during Zygotene. It is called Synapsis. It is the first essential phenomenon of meiosis. This pairing is highly specific exact point to point pairing takes place. But this pairing has no definite starting point. Each paired (but not fused) complex structure is called as bivalent or tetrad.

• Pachytene :- (Thickening Stage).

Synapsis also initiates Crossing over. The non-Sister Chromatids of homologous exchange DNA segments during Crossing over. This process redistributes genetic information among the paired homologous Chromosome's. Crossing over produces new Combination of genes on the various Chromatids in homologous pairs. Thus, each Chromatids end's with new Combination of gene's, Crossing over is the

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form of genetic recombination & is a major source of genetic variation in a population of Specie.

- **Diplotene:-** (Duplicating stage).
  The homologous Chromosome's repel each other thus the tetrad open's length wise.

- **Diakinesis:-** (Moving Apart Stage).
  The Chiasmata between the homologous Chromosome's are dissoved Now the homologous Chromosome's spread in the entire Cell. The point at which paired Chromosomes remain in contact during first phase of Meiosis I & At which Crossing over & exchange of Genetic material occure b/w the Strands.

- **Metaphase I :-**
  The nuclear membrane disappears at the biggining of this stage or phase. Spindle fibers originate. They attach to kinetochore of homologous Chromosome from each pole. They arrange the bivalents at the equator.

- **Anaphase. I :-**
  The Kinobchore fibers Contract & the spindle or pole fibers elongate. It pull the individual Chromosome

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towards their respective poles. But the sister chromatids are not separated. But the sister chromatids are separated in Anaphase of Mitosis. Each pole receives half of total number of Chromosomes, So, this is actually a reduction phase.

• Telophase I :-

Nuclear membrane organizes around each set of 2 pole's. The nucleoli reappear. This, 2 nuclie are formed. Each nucleus half number of Chromosomes. The Division of Cytoplasm (Cytokinesis), divides the Cell into 2. It terminate's the first meiotic division. The Chromosome's decondense during this stage.

• Meiosis II .

There is small interphase in 2 daughter Cell's after the Telophase I. But replication of Chromosome's doe's not takes place during this interphase
Meiosis 2 is divided into Sub-Stage's which are just like stage's of Mitosis.

- **Prophase II :-** The Chromosome condense & mitotic apparatus is appeared.
- **Metaphase II :-** The Chromosomes are arranged at the equator.

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- **Anaphase II :-** The sister chromatid's of the Chromosome's move towards the poles.
- **Telophase II :-** The Chromosome decondence.
- **Cytokinesis :-** Cytokinesis takes place. Thus each cell divide's & form's four haploid daughter cells.

Importance of Meiosis.

- **Variation :-** 2 significant processes take place during meiosis. which causes variation.
- **Crossing over. :-** Parental Chromosomes exchange segment's with each other during Crossing over. It results in large number of recombination.
- **Random Assortment of Chromosomes :-** The separation of homologous Chromosome i's random during Anaphase. It give's large variety of gametes.

⇒ These variation's are the base of evolution.

⇒ These variation's makes the animal unique & powerful in his characteristics.

Meiosis ensure or maintain Chromosome number Consent in generation after Generation in Sexually reproducing Specie.

Do You Know?

- The complete meiosis process in Human male's takes about 74 hour's.
- Meiosis was discovered for 1st time in Sea Urchin eggs in 1876 by German biologist Oscar.

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Enzyme's

"Enzymes are included in Both (CSS, PMS)"

Enzyme's & their Characteristics.

- **Dafination.**

Enzyme's are globular protein's which are biological active.

Characteristics.

- **Globular proteins:-**

All are Globular Proteins.

- **Increase the rate of reaction:-**

They increase the rate of reaction. However they are not used up in a chemical reaction.

- **No effect on end product:-**

Their presence does not effect the nature or properties of end products.

- **Specificity:-**

They are very specific in their reaction e.g Cellulase enzyme only digest Cellulose.

- **Sensitivity:-**

They are very sensitive to small change in.....

- pH
- Temperature.
- Substrate Concentration.

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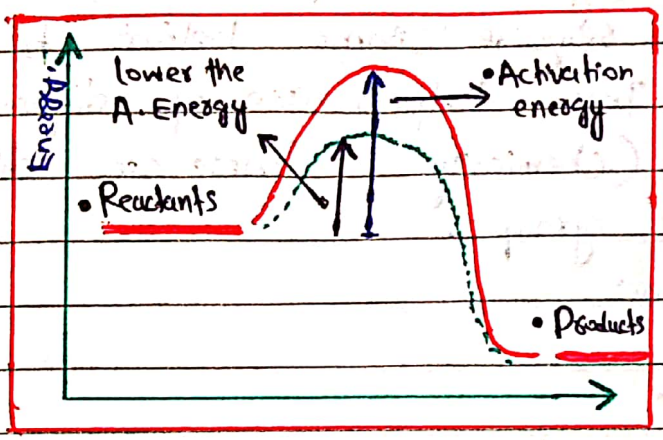
- **Activation energy :-**

They lower the activation energy. (minimum amount of energy required to start a reaction)



- figure shows that enzyme (catalyst) lowers the Activation energy.

- **Damaging effect of Enzymes..**

Some enzyme's are damaging if they become active at wrong place. Pepsin is an Protein digesting enzyme. It can destroy cells internal structure. So thats why it is produced in inactive form by lysosomes.

- **Substrate & product.**

The substance on which the enzyme acts is called reactant or Substrate. The new material formed during enzyme action called product.

- **Do You Know ?**

Single enzyme molecule can breakdown 1000 molecules of Substrate in one Second.

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Co-factor & Co-enzyme's

Some enzymes have ability to act immediately & independently. Some enzyme require some helpers to become active. These helpers are of different nature (chemically). These are divided into 2 groups.

- Co-factors.
- Co-enzymes.

Cofactors.

Cofactors are metal ion's Such as Ca^{+2}, Mg^{+2}, Cu^{-2} & Zn^{-2}.

Importance.

Cofactor Changes the Shape of Active site. It allow a better fit between the enzymes & Substrate.

Many enzyme's use these metal ion's to change non-functional active site to functional one. In these enzymes the attachment of Cofactor, changes the Shape of protein (Apoenzyme). It allows it to combine with Substrate.

The Cofactor of other enzyme's participate in the temporary bonds b/w the enzyme & its Substrate when the enzyme Substrate is formed.

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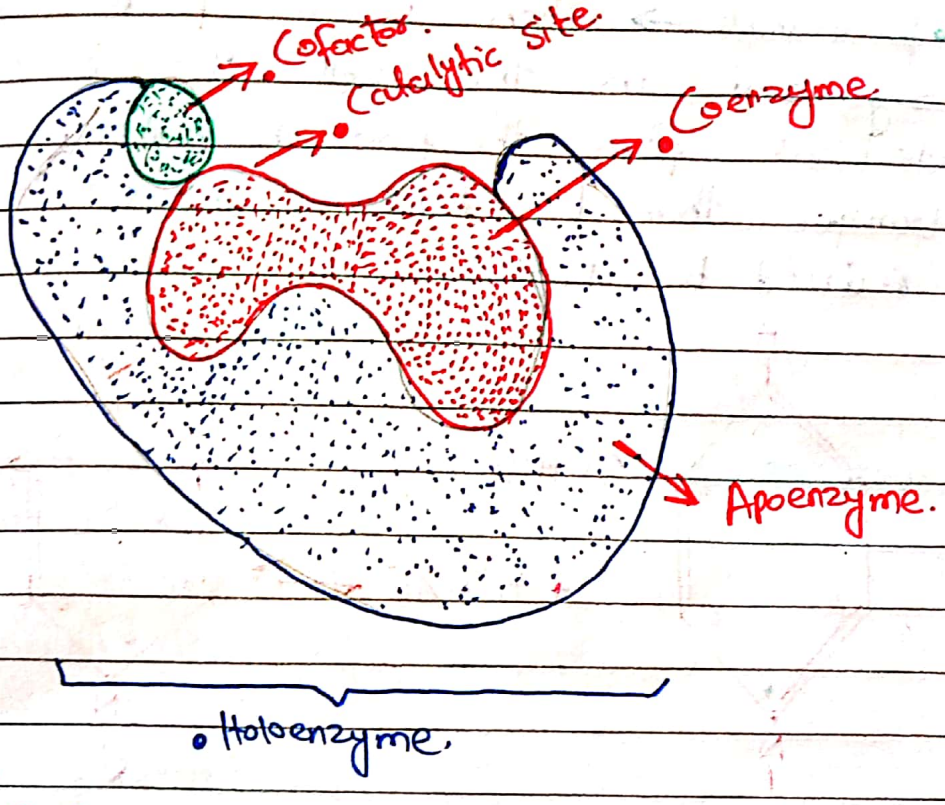

Co-enzyme.

Co-enzyme are non-protein, organic molecule's that part in enzyme-Catalyzed reaction, often by transporting electron in the form of hydrogen atom's from one enzyme to another.

Importance.

Most important Co-enzyme in the cell is the hydrogen acceptor Nicotinamide adenin dinucleotide. (NAD^+). It is formed from Vitamen B (Niacin or B_3). When NAD^+ acquires a hydrogen atom from an enzyme it reduces

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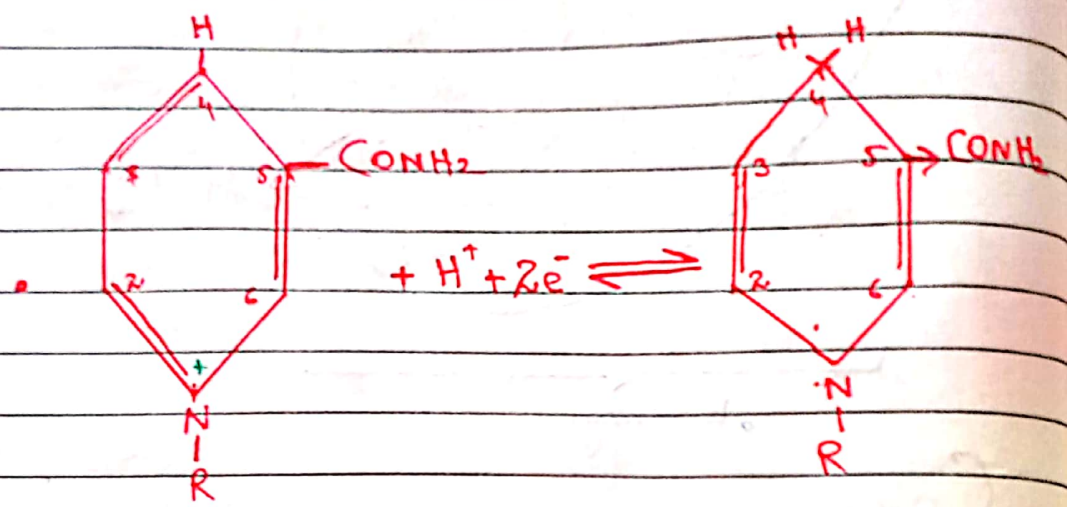
to NADH. The electron of hydrogen atom Contains energy that NADH molecule then Carries.

- for-example → When various foods are organized in the Cell the cell removes electron's from the food molecules. It transfer them to NAD. It is reduced to NADH.



- NAD.
- Oxidized form.

- NADH
- Reduced form.

- When NAD+ is reduced one electron attaches to the nitrogen atom at position 1. The second electron attaches to a Carbon atom at position 4 it is accompanied by a hydrogen ion.

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(CSS, pms)

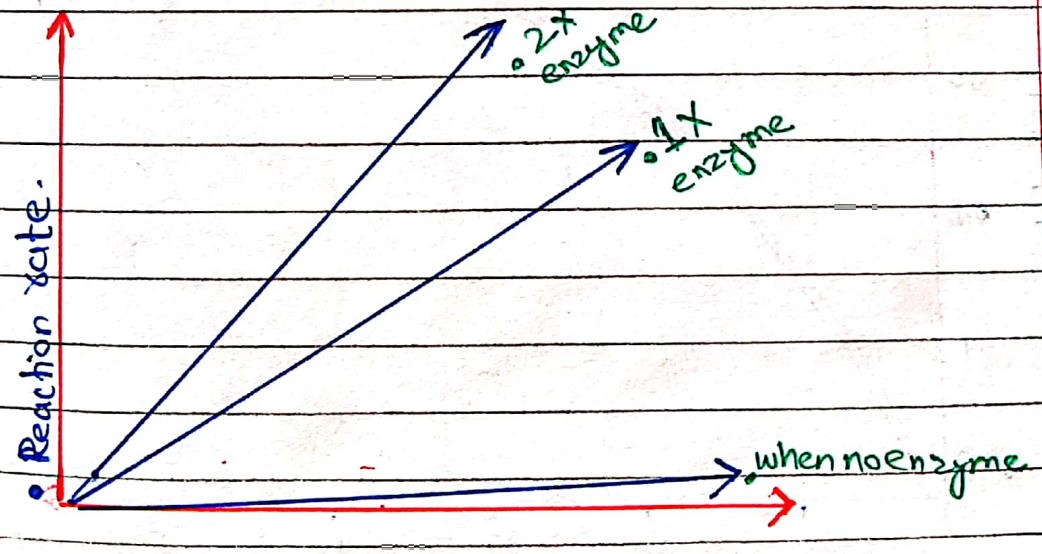
Factors Affecting the rate of Enzym Action:

The functional Specificity of every enzyme is the Consequance of its specific Chemistery & Configuration. Any factor that can alter the Chemistery & Shape of an enzyme. can affect its rate of Catalysis. Some of the important factor's that effect the rate of enzyme action are the following..

- Concentration of enzyme.
- Concentration of Substrate.
- Temperature
- pH of medium.

Concentration of enzyme.

The rate of reaction depends directly on amount of enzyme present at a specific time at unlimited amount of Substrate. By increasing



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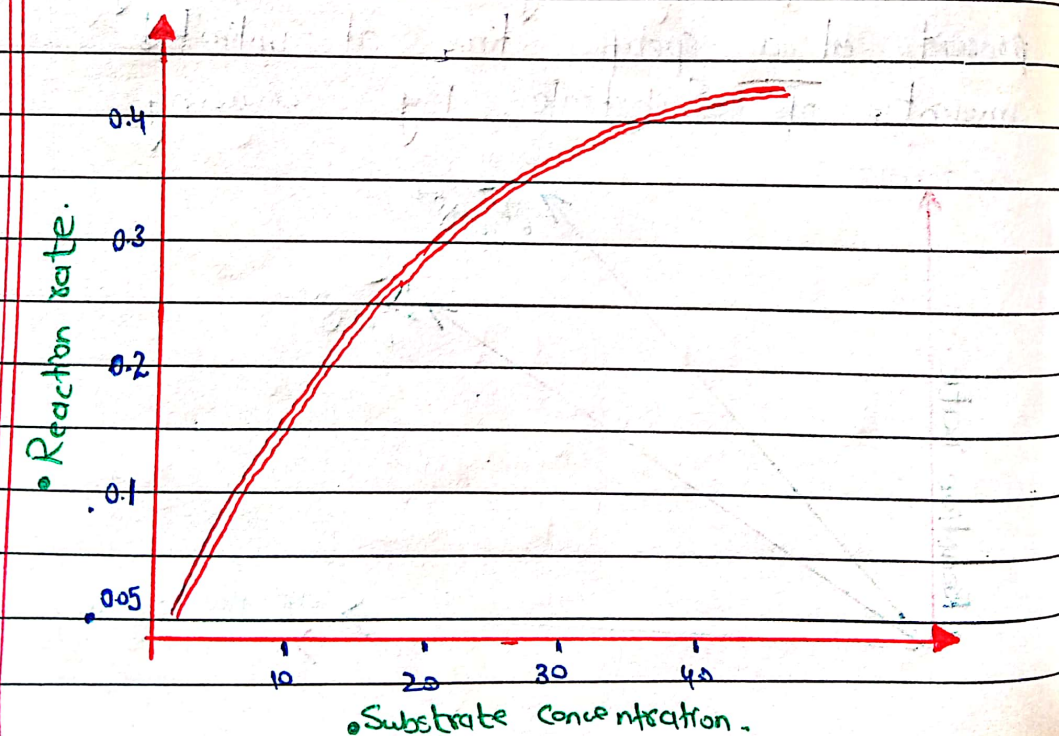
the rate of reaction ~~amount of enzyme~~ the rate of reaction is also increased because active sites are increased. More active sites will convert more substrate molecules into products in a given period.

• Concentration of Substrate. •

At low Concentration of Substrate the reaction rate is directly proportional to Substrate available. At Constent concentration of enzyme an increase in amount of substrate increases the rate of reaction. But a point is reached when no increase in rate of reaction occurs. The reason is that at high substrate level all active site's are occupied. A further increase in substrate does not increase the rate of reaction.



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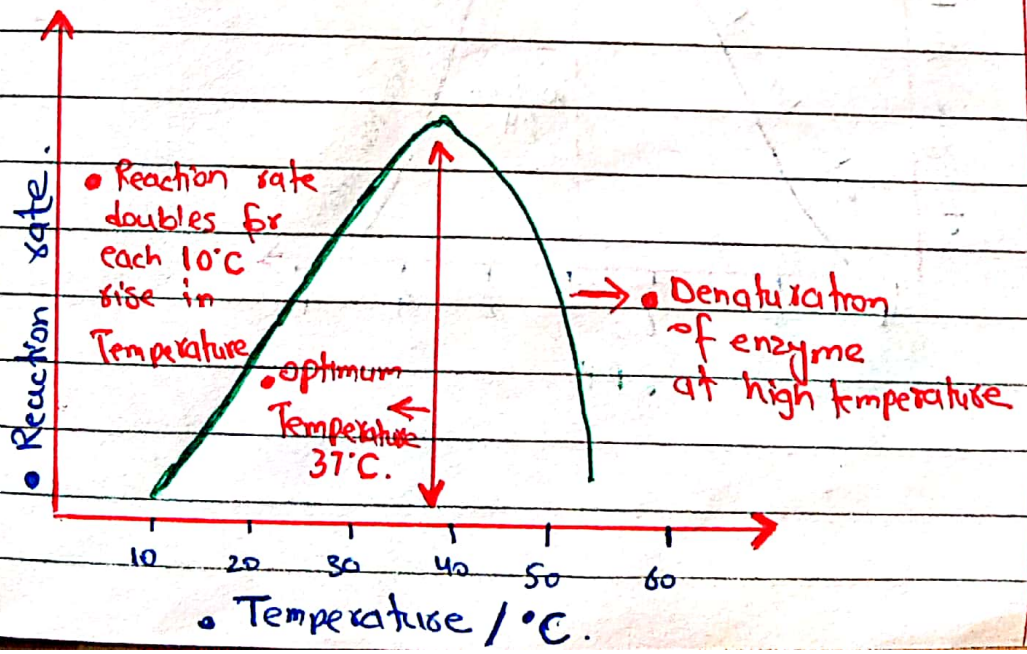
Temperature.

The shape of nuclic acid or protein is determined by hydrogen bond. Temperature change's easily disrupt hydrogen bond's, Thus changing the structure of the enzyme.

Higher Vertebrate's such as birds & mammals have enzyme's that show maximum function within a relatively narrow temperature range b/w 35 – 40°C. known as Optimum Temperature.

- Effects due Change in Optimum Temp.
- **Below 35°C :-** In mammals temperature of below 35°C, the bond's that determine enzyme's shape are not flexible to permit the shape Change necessary for Substrate to fit into a reactive site.
- **Above 40°C :-** At temperature of Above 40°C. the bond's are too weak to hold the enzyme in proper positioning & to maintain its shape.



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When due to power shape is lost the enzyme is destroyed. This loss of shape is called denaturation.

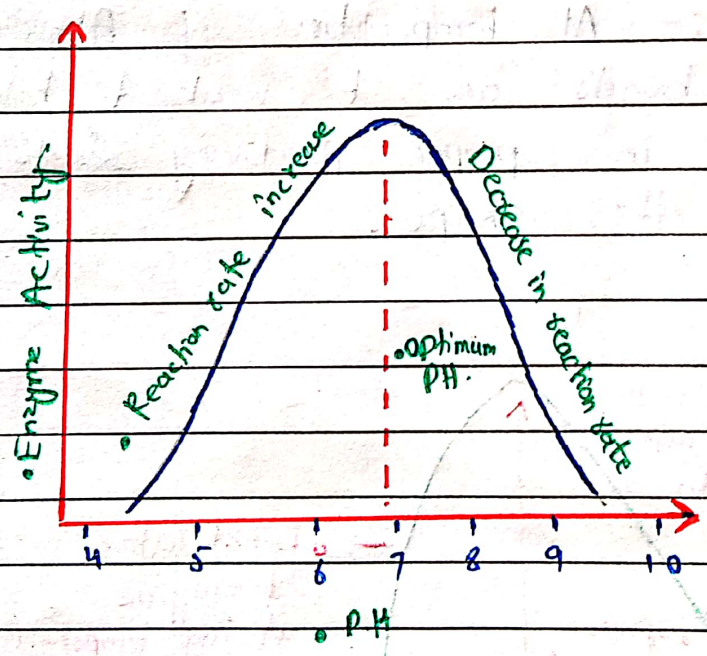
• pH.

pH is the negative logarithem of hydrogen ion concentration.

Most enzymes have a pH optimum usally between 6 to 8.

• At low pH.

When the pH is too low the H^+ ion's combine with the R-group of the enzyme's amino Acids, reducing their ability to bind with Substrate.



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Explain Inhibitors. (CSS, PMS)

- Dafination.

An inhibitor is a chemical substance which can react (in place of Substrate) with the enzyme but is not transformed into product's (s).
Thus it blocks the active site temporarily or permanently.

- Examples -

Poisons like Cynide, Antibiotics
Some drug's acts as inhibitor.

Types of Inhibitors.

Inhibitor's are divided into 2 types.

- Irreversible inhibitors.
- Reversible inhibitors.

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Irreversible inhibitors.

They check the reaction rate by occupying active sites or destroying the globular structure. They occupy the active sites by forming Covalent bonds or they may physically block the active site.

> **Do you know?** Pencillin is an irreversible inhibitor which inhibit the enzyme transpeptidase. So the growth of the bacterial cell wall is inhibited.

Reversible inhibitors.

Reversible inhibitors form weak linkages with the enzyme. their effect can be neutralized completely or partially by an increase in Substrate Concentration.

→ Reversible inhibitors are further divided into 2 major types.

Competitive inhibitors.

It is an inhibitor that resembles the normal substrate binds to the enzyme, usally at the active site & prevents the substrate from binding.

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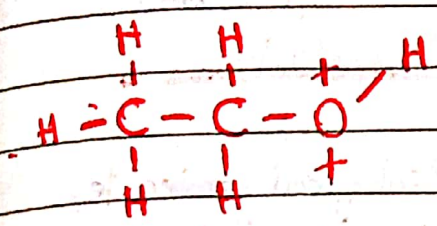
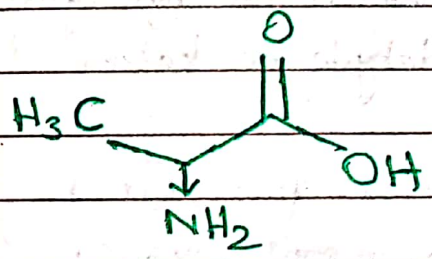


(C2H5OH)
• ethanol.

• Ethanol serves as a Competitive inhibitor to methanol & ethylene glycol for the enzyme Alcohal dehydrogenase in the liver when present in large amount.

• Non- Competitive inhibitors

These inhibitors form enzyme inhibitor complex at a point other than active site. They alter the structure of enzyme in such way that even if genuine substrate binds the active site, Catalysis fail's to take place.



• $C_3H_7NO_2$
Alanin.

• Amino Acid Alanin non- Competitively inhibits enzyme Pyruvate Kinase.

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(CSS, PPSC)

Introduction to Biomolecules.

→ Biomolecule also called Biological molecule, any of numerous substances that are produced by cells of living organisms. These are divided into 4 major types.

- Carbohydrates.
- Lipids.
- proteins.
- Nucleic Acids.

(Biomolecules are only included in CSS).

Carbohydrates.

Defination.

Carbohydrate's are polyhydroxy aldehydes or ketones which on hydrolysis produce polyhydroxy aldehyde or ketone Sub-unit's.

Carbohydrate's are derived from Greek word Sarck which mean's Sugar.

Chemical Composition.

The word carbohydrates literally mean's hydrated Carbon's. They are composed of Carbon, hydrogen & oxygen. The Ration of hydrogen & oxygen is the Same as in water. General fomula $C_x(H_2O)_y$.

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- Occurance.
  - Carbohydrate's are abundent in living organisms. They are fund in all organisms & almost all part's of the Cell.
- Source.
  - Main source of Carbohydrate's are green plant's. These are primary product of photosynthesis.
- Examples.
  - Cellulose. present in wood, Cotton & paper.
  - Starche's present in Cereal's, root tuber's e.t.c.
  - Cane Sugar. present in milk sugar & glucose.

Classification of Carbohydrates.

Carbohydrate's are Classified into 3 types.

- Monosaccharides.
- Disaccharides.
- Polysaccharides.

Monosaccharides.

- Properties.
  - These are simple sugar
  - These are sweet in taste.
  - These are easily soluble in water.
  - Canot be hydrolyzed in S. Sugar.
- Chemal Nature.

They are either polyhydroxy

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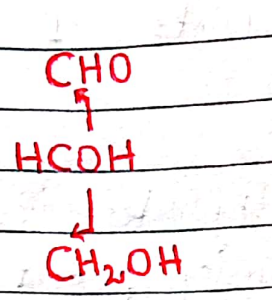
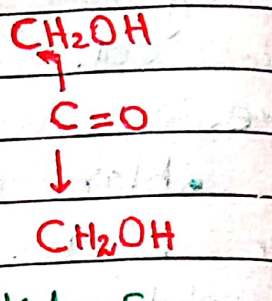
aldehydes or ketones.
The sugar with aldehyde group is called aldo-sugar & with keto sugar.



- Aldehyde form
- Glyceraldehyde.



- Keto form
- dihydroxyacetone.

Monosaccharides. Classification

⇒ As we know that monosaccharides are simple sugar's in which there are one oxygen atom & 2 hydrogen atoms for each carbon atom present in the molecule.

Natural monosaccharide's with 3 to 7 Carbon atoms are found. which are the following.

- **Trioses :-**
  Contain 3 Carbon's. e.g dihydroxy acetone & glyceraldehyde.
- **Tetroses :-**
  Contain 4 Carbon's. rare in nature present in some bacteria.
- **Pentoses :-**
  Contain 5 carbon's, most Common eg. Ribose.

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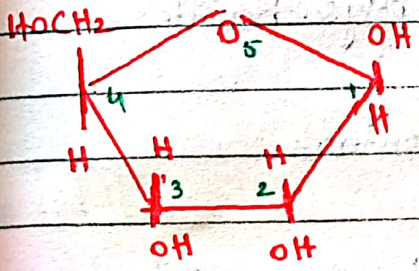
- Hexoses :-

Contain 6 Carbon's, It is aldose sugar e.g. Glucose.

- Monosaccharide's in Solution.

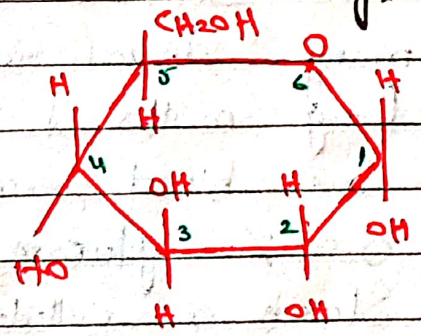
- Most monosaccharide's form ring in solution.

- Ribose form 5 Cornered ring.



- Ribofuranose.

- Glucose form 6 Cornered ring.



- Glycopyranose.

Glucose

Glucose is an important monosaccharide.

- Glucose production

Green plants take CO_2 from air & H_2O from soil to make glucose. Energy is consumed in this process which is provided by Sunlight there for this process is called photosynthesis.

Do You Know?

For the synthesis of only 1g gram of glucose 717.6 Kcal of solar energy is used.

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$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{light energy}} C_6H_{12}O_6 + 6O_2 + 6H_2O$

Presence of Glucose...

- In free state Glucose is present in about all fruits. But it is abundant in grapes, figs & dates.
- In Combined form it is present in many disaccharides & polysaccharides. (e.g. maltos & Glycogen).
- Our blood normally Contains 0.08% glusose.
- Starch, cellulose & glycogen produce glucose on hydrolysis.

Disaccharides...

Two monosaccharide's combine to form disaccharide's by removing a molecule of water. (dehydration synthesis).

Disaccharide's as Isomers...

All disaccharides have same molecular formula $C_{12}H_{22}O_{11}$. Such Compaind's with the same molecular formula but different Structure's are called Isomer's. Each Isomer has unique properties.

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Example's of Disaccharide's.

- **Sucrose :-**

Disaccharid formed by the linking of Glucose molecule to a fructose molecule.

- Glucose + fructose ⟶ Sucrose.

- **Lactose :-**

A glucose molecule bond's to glactose to form lactose.

Glucose + Galactose ⟶ Lactose.

- **Maltose :-**

It is formed by joining of 2 glucose Subunit's. It is used in alcohal.

Glucose + Glucose ⟶ Maltose.

Polysaccharides.

Polysaccharides are long chain's of monosaccharide's linked by Glycosidic bond. Glycosidic bond or Glycosidic linkage is a type of Covalent bond that join's a Carbohydrate (Sugar) molecule to another group which may or may not be another Carbohydrate.

Characteristic's of polysaccharide's.

→ These are most Complex of most abundant Carbohydrate's in nature.

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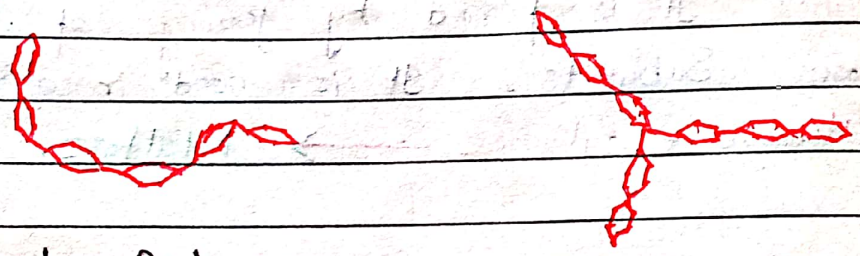
→ They are usually branched & tasteless.
→ They have high molecular weight.
→ They are springly (ﻉﺎﻨﻴ) Solve in water.

• Examples of Polysaccharide's •

• Starch :-

• Properties.
→ Main Source of Carbohydrate's for animals.
→ Gives blue Color with Iodin.
→ found in Grains, seed & tubers.

Starch is further divided into 2 types.
• Amylose Starches :- Unbranched Glucose chain Soluble in Hot water
• Amylopectin Starch :- Branched chains soluble but in water.



• Amylase Pectin
• Unbranched.

• Amylopectin
• branched

• Cellulose :-

• properties.
→ Main Component of plant Cell wall.
→ Most abundant Carbohydrate in nature.
→ Highly insoluble in water.
→ Gives no color with iodine.
→ Not digested in human digestive tract.

• Example.
• Cotton is a pure form of Cellulose.

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- **Glycogen :-**

Main form of Carbohydrates in animals.
found in all cell's but abundant in liver & Muscle Cells.
Insoluble in water.
Gives red Colour with iodine.
Mainly known as Animal Starch.

- **Chitin :-**

It is major Component of exoskeleton of insect & Crustacians such as lobsters & Crabs.

Do You know?	Do You know?
(Cellulose) is the most abundent Carbohydrate on Earth. mainly found in plant cell wall.	Most abundant Carbohydrate in human body is (Glucose) & most abundent in human blood is (Starch).

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Lipids. (CSS)

Dafination:-

Lipids are non-polar organic compounds or molecules that are insoluble in polar water but soluble in non-polar organic solvents such as ether, alcohol, & Chloloroform. e.t.c

Chemical Composition:-

Lipid molecules are composed primarily of Carbon, Hydrogen & oxygen atoms. Some may Contains small amount of phosphorus & Nitrogen. They Contain much smaller portion of oxygen than do Carbohydrates as the formula for the fat tristerean illustrate's $C_{57}H_{110}O_6$.

Classification of Lipids.

Lipids are Classified into 3 forms

- fats.
- phospholipids
- Steroids.

fats. (1)

fats are ester of fatty acids & glycerol.

- Building Blocks of fat Molecules.

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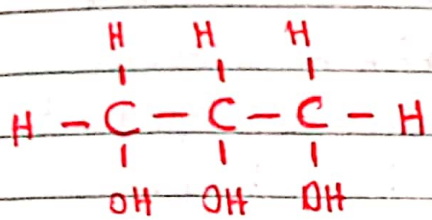
The building blocks for fat molecules are.....

- **Fatty Acids:-**

fatty acid Contain long hydrocarbon chains bonded to Carboxyl (-COOH) groups.

- **Glycerol:-**

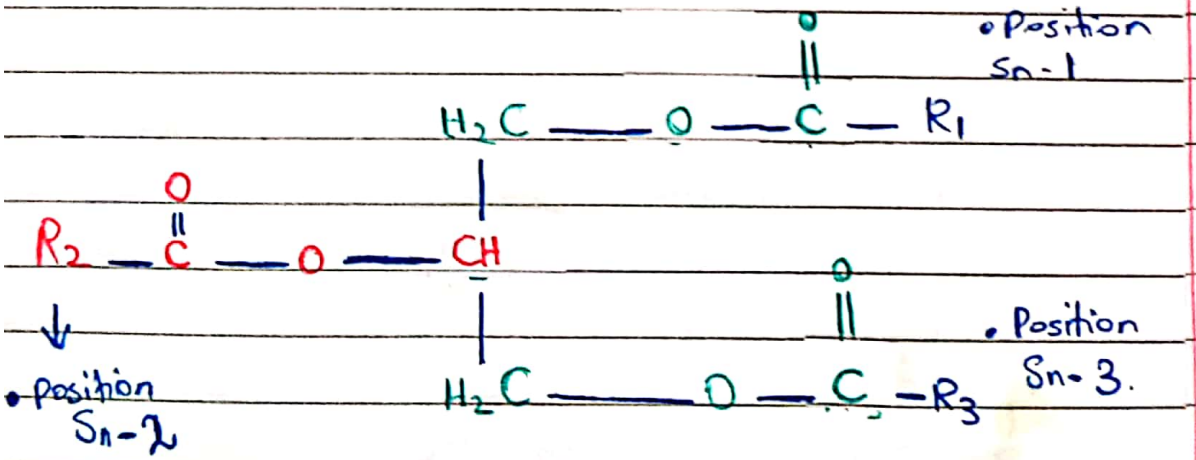
Glycerol is the 3 Carbon alcohol with each Carbon bearing a hydroxyl group (-OH).



- $C_3H_8O_3$
- Glycerol.

- **Triacylglycerol:-**

The fats with 3 fatty acids joined with glycerol are called Triglyceride neutral fat or triacylglycerol.



- General Structure of triacylglycerol.

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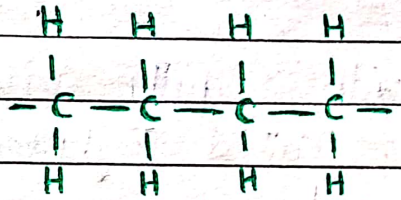
Types of fats.

Saturated fats.

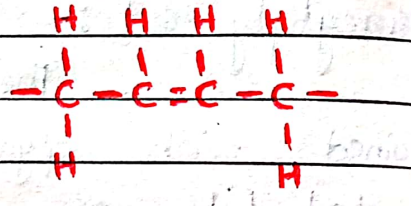
- All the Carbon atoms in the chain are linked by single bond & bear hydrogen atoms.
- They cannot takeup more hydrogen.
- These are straight Chain molecules.
- They are solid at ordinary temperature.
- They are less abundant in living organisms
- present in most animal fats.



- Saturated.

Unsaturated fats.

- All the Carbon atoms in the chain are linked by double bonds & Cannot bear H. atoms.
- They can Takeup more hydrogen.
- They are branched Chain molecules.
- They are liquid at ordinary temp.
- They are more abundent in living organisms
- Occure in most plant fats.



- Unsaturated.

Phospholipids.

Phospholipids are known as phosphatides Consider the Class of lipids whose molecules has a hydrophilic

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head Containing a phosphate group & 2 hydrophobic tails derived from fatty acids joined by Glycerol molecule.

• Parts of phospholipids.

It has 2 parts .....

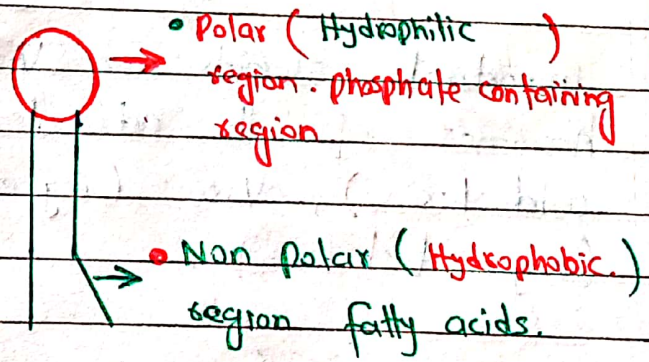
- Head
- Tail

• Head.

The polar phosphate & nitrogen groups are soluble in water (Hydrophilic) & form the head of the molecule.

• Tail.

fatty acid portion of molecule forms the tail which insoluble in water Hydrophobic.



• Steroids.

Steroids are naturally occuring lipid soluble molecules.

• Properties of Steroids.

- They are composed of 4 fused

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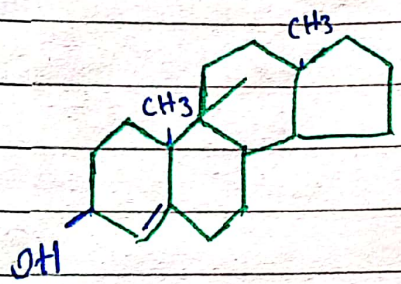
Carbon rings.

- It forms a rigid structure.
- 3 of rings are 6 sided & 4th is 5 sided.
- The 4 rings contain total of 17 Carbons.

- **Important Steroid examples.**

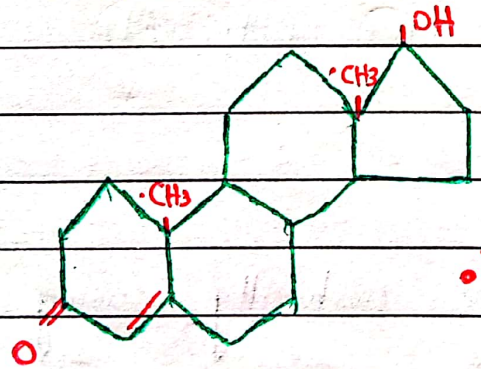
- **Cholesterol.**

It is an important biologically active steroid.



- **Vitamin D & Hormones.**

Similarly some important steroids are Vitamin D & hormones of the adrenal gland (eg. aldosterone), ovaries (e.g estrogen) & testes (e.g. testosterone).



- Testosterone steroid.

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• Do U You?

→ Steroids which are the most abundant on Earth are "Sterols" which occure in all tissues of animals. green plants & fungi.

• Do You Know?

Phospholipids are most abundent lipids which makes about 70%. of membrane lipids.

• Importants functions of lipids.

→ phospholipids are found in Cell membranes of plant animal & bacterial Cell.

→ Lipids are Source of energy They can Store double the amount of energy as Compared to the same ammount of Carbohydrates.

→ Some lipids acts as water proof material (e.g. Wax in invertebrates)

→ Some lipids provide insulation against atmospheric heat or Cold. (e.g. fats in body of animals protect heat dissipetron).

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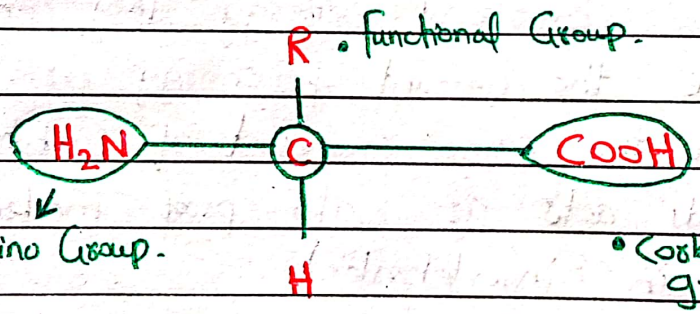
Proteins (CSS)

- **Dafination :-**
  Protein's are polymers of Amino Acid.
- **Chemical Composition :-**
  Protein's always contain atom's of Carbon, Nitrogen, Hydrogen Oxygen & Some time Sulful.

- **Amino Acids.**
- **Building blocks of proteins :-**
  The individual building blocks of protein's are amino Acid's.
- **Structure of Amino Acids :-**
  Amino Acid's alway Contain a amino group ($-NH_2$) a Carboxyl group ($-COOH$) A hydrogen at & a functional group (R), These all bonded to a central Carbon atom.



- **Kinds of Amino Acids :-**
  Twenty different amino acids Commonly occure in animals.

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Various level of Structure of proteins.

The protein molecule has saveral different level's of Structure. e.g.

- primary Structure.
- Secondary Structure.
- Terbiary Structure.
- Quaternary Structure.

Primary Structure :-

Primary Structure is the linear sequance of amino acid's in the polypeptide chain's.

Secondary Structure :-

The secondary structure of a protein is a repeating pattern of bond's (often hydrogen bonds) between amino acid's & it commonly takes the shape of Alpha helix, β pleated sheet.

α helix

- It is halical in Stucture.
- It is simple Unit
- In each helical turn 3.6 amino acid's are present.

e.g. karatin, hair's, neils

β-sheet

- It is plate like form.
- It is complex Unit.
- No such sequence is seen in it.

e.g. Silk, spider, web.

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• Tertiary Structure :-

In a tertiary structure a polypeptide chain bends & folds upon itself (اپنے اوپر بار بار لپیٹ کر) forming a globular shape (Ball like shape).

• Structural Maintenance :-

Tertiary structure is maintained by 3 types of bonds.

- Ionic bonds
- Hydrogen bonds
- Disulfide (S-S) bonds.

• Quaternary Structure :-

In Quaternary protein's are polymers of several tertiary structures:
In this structure, the highly complex polypeptide tertiary chains are aggregated & held together by....

- Hydrophobic interaction's
- Hydrogen bonds
- Ionic bonds.

Do You Know?
• Rubisco is the most abundent protein on Earth. responsible for Global Carbon fixation

Do Yo Know.
Collagen is the most abundent protein in humen body found in bones, muscles & skin.

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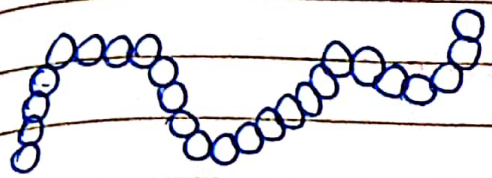
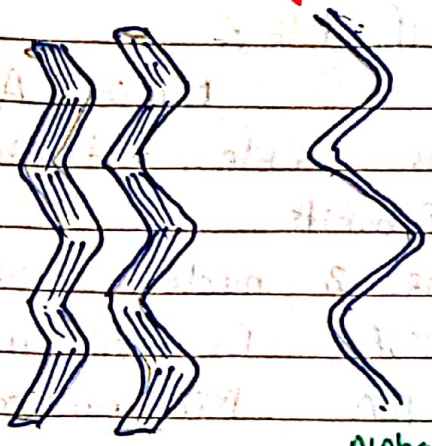
- Primary Structure.



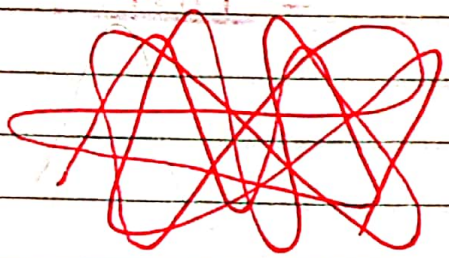
- Secondary Structure.



- Beta. Alpha.

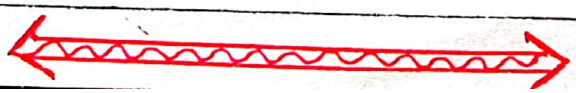
- Tertiary Structure.

- Quaternary Structure.



Important functions of Proteins

- All enzymes are protein in nature. (lipase, pepsin)
- Antibodies are proteins which protect the body from pathogens.
- Movement of organs & organisms are also caused by protein's (Actin & myosin).
- Some proteins work as carrier (e.g haemoglobin transport oxygen).
- They build many structures of the Cell. e.g proteins are involved in formation of Cell membrane.



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Nucleic Acids.

- **Dafination :-**

Nucliec Acids are large single or double stranded chain's of nucleotide Subunit's

The 2 nucleic acids are.....

- DNA Deoxyribonucliec Acid.
- RNA. Ribo-nucliec Acid.

:- DNA :-

- **Dafination :-**

DNA or deoxyribonucliec acid is a genetic material that transfer the genetic information from one generation to their offspring.

- **Location :-**

located in Nucleus or mitochondria.

- **Shape or Structure...**

→ DNA is a long chain polymer of nucleotide's.

- Consisf of.....
  - → Deoxyribose = 5 pentose Sugar
  - → Phosphate group
  - → Organic basis,
    - Adenin, Cuanine (pusines)

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- Cytosine , thymine (pyrimidines).

Application of DNA :-

- DNA finger printing.
- DNA Sequencing
- Recombinent DNA tecnology.

Function of DNA :-

- Replication
- Encoding information.
- Mutation / Recombination.
- Gene expression.

- Do You Know ?
  - Your DNA can stretch from earth to the Sun & back 600 time's.
  - Our (human genome) genome contains 3 billion base pair of DNA.
  - Human Share 99% of their DNA with Chimpanzee.

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RNA

Defination.

It is an Abbreviation of Ribonucliec Acid. It is an polymeric molecule essential in various biological roles in Coding, decoding, regulation & expression of genes.

Location.

Mostly found in nucleolus & then later move's to the specialized region's of the Cytoplasm depending on the type of RNA formed.

Structure.

RNA Consists of 4 nitrogenous base's
Adenin.
Cytocine
Uracil
& Guanine.

- RNA is a typically single stranded & is made of ribonucleotide's that are linked by phosphodiester bond's.
- RNA contains Sugar ribose.

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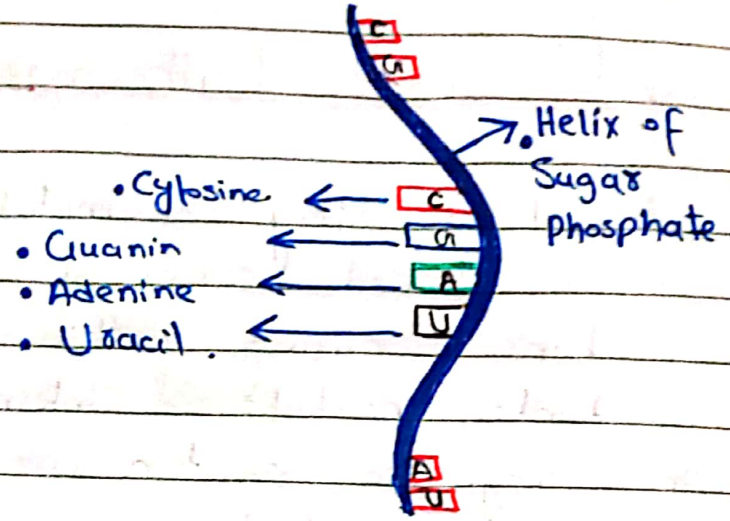
• RNA Stoachure



• Functions & type's of RNA

• MRNA :- Called mesenger RNA it contains informations which are going to translate into protein.

• tRNA :- Called transfer RNA it carries & bind's Amino Acid's as an Anticodon.

• rRNA :- Called ribosomal RNA help & facilitate the banding of amino Acid's Coded fr by mRNA.

• Do You Know?

• Abaut 5% of the weight of a human cell is RNA & only 1% of Cell consist of DNA.

RNA is the first molecule of heridity because Prokaryote's have RNA as their genetic material.

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Metabolic Pathways :-

In biochemistery a metabolic pathway is a linked series of chemical reaction's occuring within the cell. The reactants, products & intermediate's of an enzymatic reaction are known as metabolite's. which are modified by a sequance of chemical reaction's catalysed by enzymes.

There are 3 main kind's of metabolic pathways.

- Glycolysis.
- Kreb's Cycle.
- Electron transpost chain.

Explanation

Glycolysis. (CSS, PMS)

- Dafination

Glycolysis is the first step of Catabolic chemical reactions. It occurs in allmost all kind of Cell's. Six Carbon glucose molecule is broken down into 2 molecule's pyruvate during glycolysis. It is also known as Anaerobic pathway or (Embden Meyerhof pathway).

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Pyruvate :-

Pyruvate is a 3 Carbon compound It is the output of metabolism of glucose Known as glycolysis.

Phase's in Glycolysis :-

There are 2 main phase in Glycolysis.

- Energy Investment Phase.
- Energy Yielding phase.

Energy Investment phase :-

There are 5 Step's in energy investment phase.

Step 1 :-

Glucose enter's into the cell ATP used & It is phosphorylated by Hexokinase. Phosphorylation of glucose make's the molecule more chemically reactive.

Step 2 :-

Glucose 6- phosphate is rearranged by enzyme Isomerase. It is converted into its isomer. fructose -6- phosphate.

Step 3 :-

In this step another molecule of ATP is used in Glycolysis. & the reaction is catalyzed by. Phosphofructokinase.

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- **Step 4 :-**

In this step Splitting of fructose 1,6 biphosphate is Catalyzed by enzyme Aldolase. to form 2- Three Carbon products which are Dihydroxyacetone phosphate & Glyceraldehyde 3- phosphate.

- **Step 5 :-**

This is the last step of energy investment phase. Dihydroxy- acetone phosphate is isomerized by Triose phosphate isomerase to form Glyceraldehyde 3 phosphate. & now only Glyceraldehyde 3 phosphate will be used in the next Step.

- **How to remember Glycolysis Enzymes.**
- Hamza proposed Paenda at Turkey But Purwez proposed Esha in Pakistan.

- Hamza (H) ⟶ Hexokinase.
- proposed (P) ⟶ phosphoglucose Isomerase.
- Paenda (P) ⟶ Phospho-fructo kinase
- At (A) ⟶ Aldolase.
- Turkey (T) ⟶ Triose-phosphate dehydroge...
- Purwez (P) ⟶ Phosphoglycerate kinase.
- proposed (P) ⟶ Phosphoglycero mutase.
- Esha. (E) ⟶ Enolase.
- Pakistan (P) ⟶ pyruvate kinase.

- **How to remember Glycolysis Enzymes :-**

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Name of molecules		Enzymes
Glucose		
ATP → ADP		Hexokinase
Glucose-6-phosphate		
		Phosphoglucose isomerase
Fructose 6-phosphate		
ATP → ADP		Phospho fructo kinase
Fructose 1,6 bisphosphate		
		Aldolase
Glyceraldehyde phosphate + Dihydroxy acetone phosphate	NAD → NADH	Triose phosphate dehydrogenase
1,3 Diphosphoglycerate	ADP → ATP	Phosphoglycerate kinase
3-phosphoglycerate		Phosphoglycero mutase
2-phosphoglycerate	H_2O	Enolase
Phosphoenol pyruvate	ADP → ATP	Pyruvate kinase
Pyruvate		

No further reaction.

end product of Glycolysis.

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Energy Yielding Phase.

This phase starts from the 6th Step of Glycolysis.

Step 6 :-

An enzyme now catalyzes 2 reactions. first Glycealdehyde phosphate is oxidized by transfer of electrons & H to NAD. It form's NADH'. This reaction is very exergonic. It form's 1,3 bisphosphoglyceric acid The source of the phosphate is inorganic inorganic phosphate. It is always present in Cytosol.

Step 7 :-

This step is the transfer of phosphate group from 1,3 bisphospho glycerate to ADP forms 3, phospho glycerate which is catalyzed by enzyme Phosphoglycerate kinase. & it produces ATP.

Step 8 :-

In Step 8, an enzyme (mutase) relocates the remaining phosphate group. It form's 2-phosphglycerate. It prepare's the Substrate for next reaction.

Step 9 :-

In this step there is a loss of water to form phosphenol pyruvate which is catalyzed by Enolase.

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• Step 10 :-

In this step An enzyme transfers the phosphate group from phosphenol pyruvate (PEP) to (ADP). It produces more ATP Since this step occurs twice for each glucose molecule, so, there is a net gain of 2 ATP molecule. Glucose has been broken down & oxidized to 2 molecule's of pyruvate. It is end product of glycolytic pathway.

• How to remember Glycolysis Steps.

God Gives fruit for Grand-parents But Parent's Puts Pepsi pressurely.

- God → G → Glucose
- Gives → G → Glucose-6-phosphate
- fruit → F → Fructose-6-Phosphate
- for → f → fructose, 1-6 bisphosphate
- Grand parents → G.p → Glyceraldehyd phosphate.
- But → B → 1,3, bisphosphate
- Parent's → P → 3-Phosphoglycerate
- Puts → P → 2-phospoglycerate
- Pepsi → P → Phosphoenol pyruvate.
- Pressurely → P → pyruvate.

• written by.
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03081127642

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Krebs Cycle (CSS, PMS)

- **Dafination:-**

It is a series of reaction in which pyruvate from glycolysis is oxidized to CO_2. This Cycle was discovered by Sir Hans Krebs in 1930's. It is also known as Tricarboxylic Acid Cycle or Citric acid Cycle.

In Krebs cycle 2 electron Carriers act as hydrogen acceptors.

Nicotinamide Adenine dinucleotide (NAD)

flavin Adenin dinucleotide (FAD).

These are reduced to NADH & $FADH_2$. Pyruvate first change into Acetyle CoA. Then Acetyle CoA enters into the Cycle.

- **Steps in Krebs Cycle.**

Krebs Cycle involves 8 steps which are the following.

- **Step 1:-**

Acetyle CoA adds its 2 Carbon fragments to Oxaloacetate. oxaloacetate is a 4 Carbon Compound. The Unstable bond of acetyle CoA is broken attaches to the acetyle group. The product is 6- Carbon Citrate & this reaction is Catalyzed by enzyme Citrate Synthase.

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• Step 2. :-

A molecule of water is removed & another is added back. The net result is the Conversion of Citrate to its isomer Isocitrate. & this reaction is controlled by enzyme Aconitase.

• Step 3. :-

In this Step Substrate lose's a CO_2 molecule in the presence of enzyme Dehydrogenase. This remaining five Carbon compound is oxidized. It reduces NAD to NADH & a Ketoglutarate is formed.

• Step 4 :-

This step is catalyzed by multienzyme Complex. CO_2 is lost. The remaining 4 Carbon ~~atom's~~ Compounds is oxidized & electron's are transfered to ~~from~~ NAD to form NADH. Thus Succinyl CoA is formed.

• Step 5 :-

Substrate level phosphorylation occurs in this step. CoA is replaced by phosphate group & Succinate is formed. Phosphate group is then transfered to GDP. It forms GTP. (Guanosine triphosphate). GTP is similar to ATP.

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(10%)

- **Step 6 :-**

2 hydrogens from Succinate are transfered to form $FADH_2$. It produces fumarate.

- **Step 7:-**

Bond's in the Substrate are rearranged by addition of water & (malate) is formed.

- **Step 8:-**

In this Step another molecule of NADH is produced. It regenerate's oxaloacetate. The oxaloacetate is ready to enter into another reaction

- **How to remember Krebs Cycle Steps.**

- Can I Keep Some Sweets for my Office.

- Can → C → Citrate
- I → I → Iso-Citrate
- Keep → K → Ketoglutarate
- Some → S → Succinyl. CoA
- Sweets → S → Succinate
- for → F → Fumarate
- my → M → Malate
- Office. → O → Oxaloacetate.

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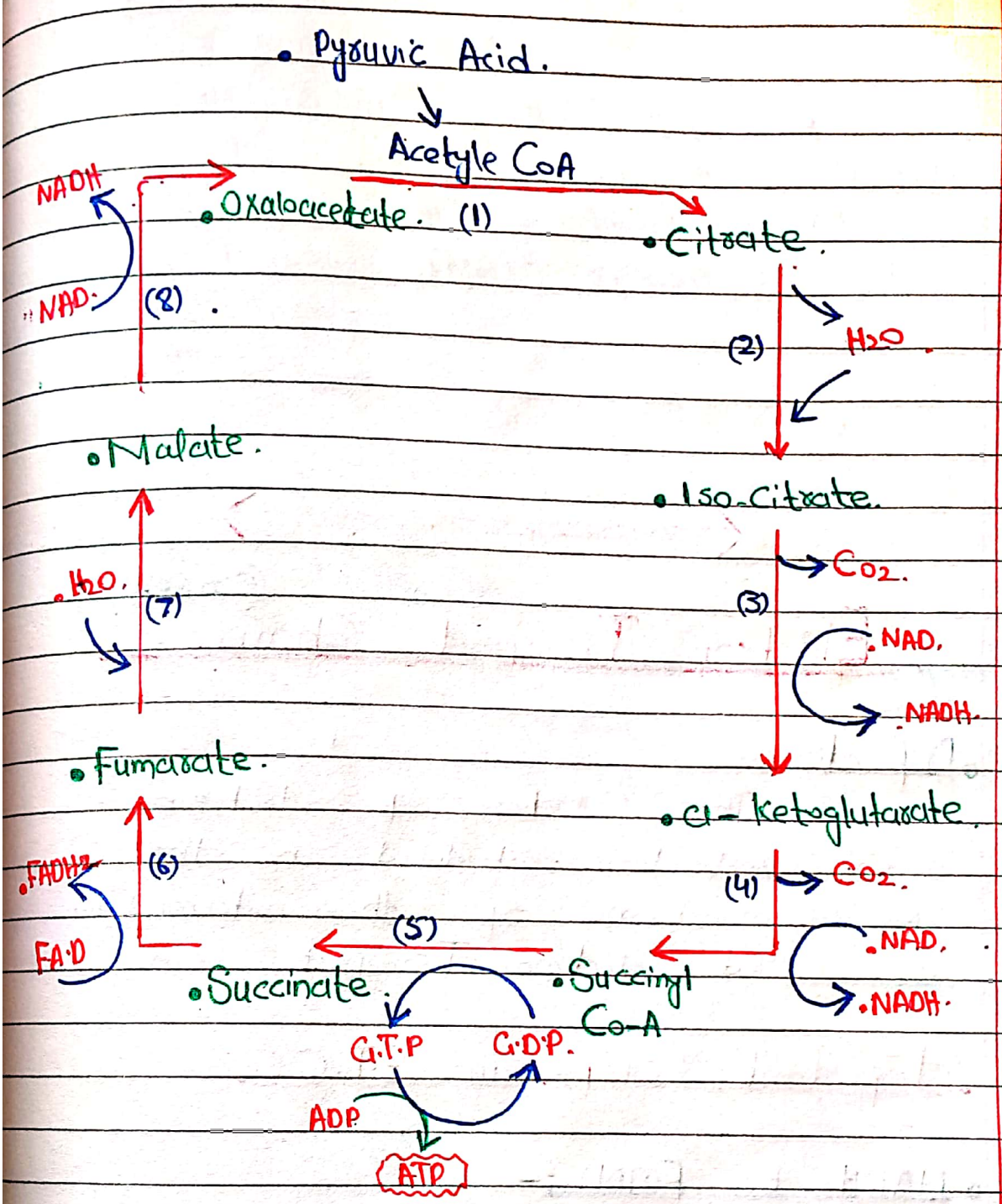

• Kreb's Cycle end product's :-

- Three molecule's of CO_2.
- Three molecule's of $NADH_2$.
- One molecule of $FADH_2$.
- One molecule of ATP.

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(CSS Pms)

Electron Transport Chain.

Dafination.

The Collection of electeron carrier molecule's embedded in the inner membrane of the mitochondria is Called electron Transport Chain.

Important Steps in E.T.C.

NAOH to FMN :-

Electeron's are removed from food during glycolysis & Krebs cycle. These electeron's form $NADH_2$. $NADH_2$ transfer electron's to first molecule of (E.T.C) flavoprotein. The flavoprotein has a prosthetic group Called mononucleotide (FMN).

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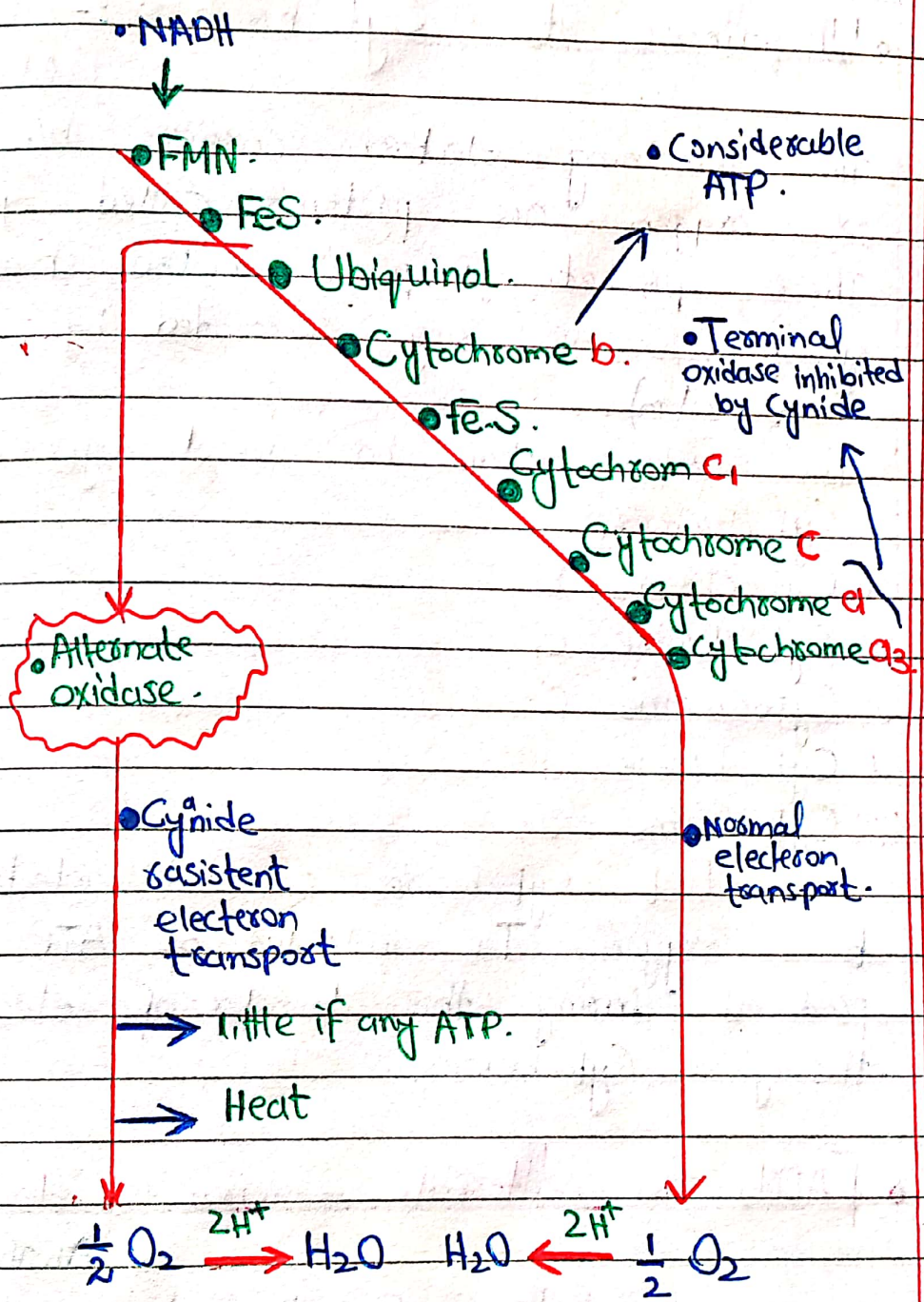
FMN to fe.S :-

The (F.M.N) passes electeron to an iron- sulfur protein (F.es) . It is one of the family protein in which both iron & Sulfur tightly bound. It is redox reaction (FMN) returns to its oxidized form..



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- **Fe.S to Ubiquinone (Q) :-**

Fe.S the passes the electron to . This electron carrier is only member of electron transport Chain that is not protein. A molecule of is Synthesized during this transfer.

- **Ubiquinone to Cytochromes :-**

Most of the remaining electeron carrier's b/w (Q) & oxygen are protein's called Cytochrome's. Their prosthetic group is called haem group. (organic ring sorrounded by single iron atom).

The electron chain has saveral types of Cytochrome's.

(a) Cytochrome b
(b) Cytochrome c,
(c) Cytochrome c
(d) Cytochrome a
(e) Cytochrome a_3

Last Cytochrome passes its electron's to oxygen. Two molecules of ATP are produced during this transfer of electron through Cytochrome's.

- **$FADH_2$ an Alternate Source of electron :-**

Another source of electron's for the transport Chain is $FADH_2$. It is produced during Krebs cycle $FADH_2$ adds its electron

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to transport chain at a lower energy level than that of NADH. Therefore, electron transport chain provide's one third less energy than $FADH_2$.
The electron transport chain makes no ATP directly. It's function is to pass electron's from food to oxygen. It breaks a large free energy drop into a series of smaller step's that release energy in manageable amount's. Mitochondria Synthesize ATP during mechanism Called Chemiosmosis.

Transcription. (CSS, PMS)

- **Dafinction :-**

The Synthesis mRNA (RNA Copy) from DNA (Gene Sequance).
The Genetic information in DNA is not directly into protein's. These information's are first transcribed in mRNA.

- **Role of enzymes in transcription.**

Many enzyme's involved in this function of process.
- Enzyme's unwind the region of DNA molecule.

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- They initiate & end mRNA synthesis.
- They also modify mRNA after the completion of transcription.

Transcription in Eukaryotes:-

Mainly 3 kind's of enzyme's are involved in transcription of different kind's of RNAs.

- RNA polymerase I :-
  located in nucleolus & transcribed ribosomal RNA (rRNA).
- RNA polymerase II :-
  located in nucleus & transcribes messenger RNA (mRNA)
- RNA polymerase III :-
  located in nucleus & transcribes transfer RNA (tRNA) & some other small RNAs (5sRNA).

Step's in Transcription:-

There are 3 main step's involved in transcription.

- Initiation
- Elongation
- Termination

Initiation :-

Concept of promotor :-

The site of DNA where DNA polymerase attaches is called promotor. promotor is mainly composed of 50 nucleotide's. It is

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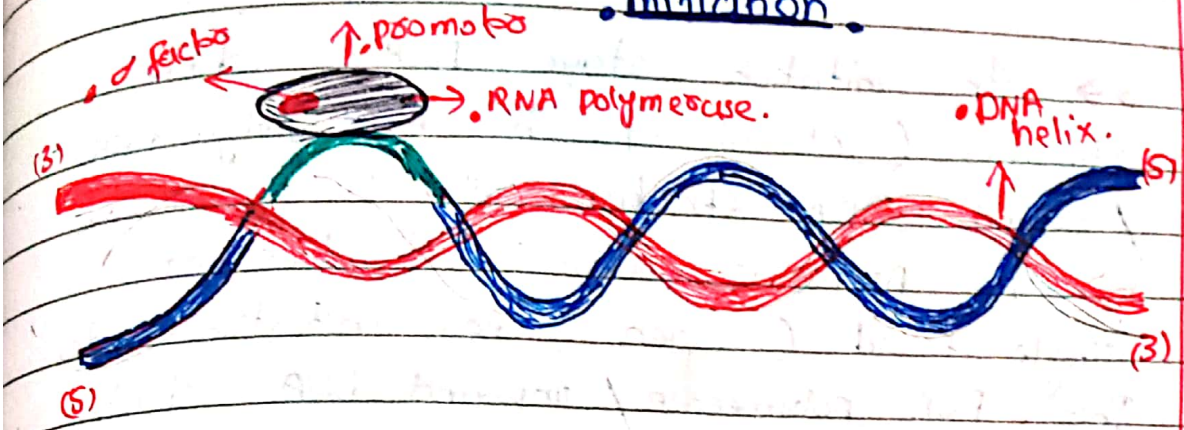
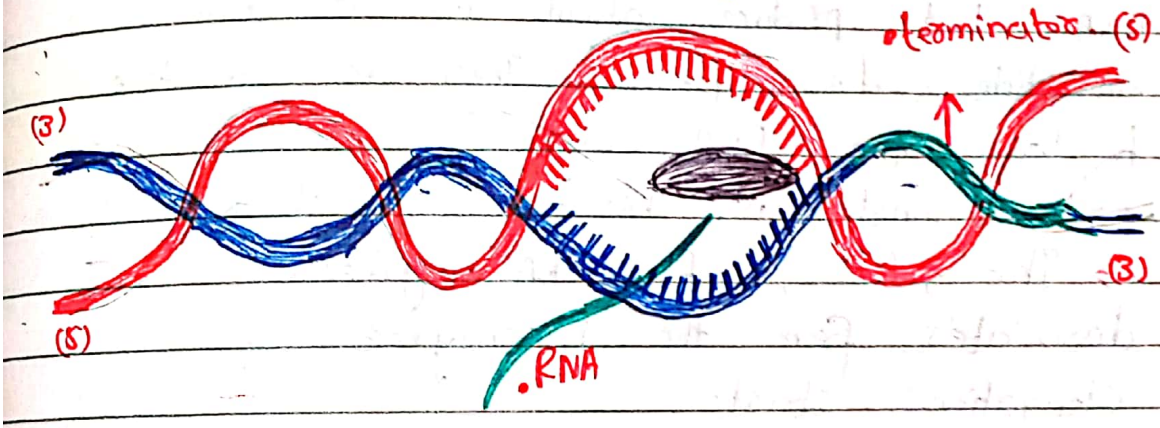
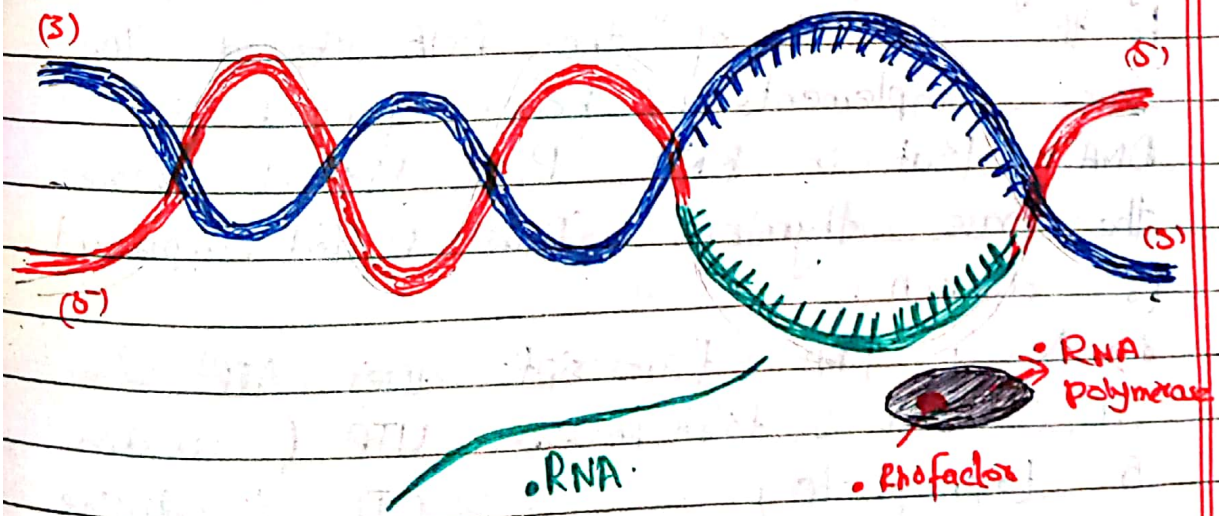
Initiation



Elongation



Termination



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present at the start of genes.

→ In initiation stage DNA is doble stranded (closed). This RNA polymerase / wound DNA structure called Closed Complex.

→ The DNA is unwound & become's single strand (open) near initiation site. This RNA polymerase / unwound DNA structure called open Complex.

→ The elongation polymerase transcribe's the DNA, but produce's about 10 abortive nucleotides (transcripts). These are unable to leave the RNA polymerase because exit channel is blocked by th (σ factor).

→ The (σ factor) of RNA polymerase dissociate's from the holoenzyme & elongation start's.

• Elongation :-

DNA polymerase starts joining complementary ribose nucleotides to the 3' end of the DNA strand. The same complementary base's paired in DNA, But in RNA. Base Uracil replace's the base thymine, Thus Uracil complement to adenin.

→ Newly RNA transcript's use's (ATP) & (GTP) Guanosine 5' triphosphate. UTP (Uridin 5' triphosphate) & CTP. (cytidine 5' triphosphate.

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Termination :-

Transcription Contineous finally RNA polymerase reaches the termination sequance.

Mechanism of termination :-

There are 2 termination mechanism's which are well known.

Rho-Independent termination :-

In Rho-independent or intrinsic termination, Rho factor bind's with transcript & terminate's transcription by breaking hydrogen bond b/w tempelate & transcript.

Rho-dependent termination :-

In Rho-dependent or termination Rho factor stop R.N.A Synthesis at specific site's. when Rho factor reaches the RNA polymerase it dissociate's. RNA polymerase from the DNA terminating transcription.

Processing of primary transcript :-

The newly transcribed mRNA is called primary transcript. It is modified before leaving the nucleus. Non. Coding regions which remain during Synthesis are removed. Thus the Coding regions of mRNA can be read Contineously at the ribosomes.

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Transport of newly Synthesized RNA

- **Prokaryotes :-** Newly synthesized mRNA is directly released into the Cytoplasm in prokaryotes.
- **Eukaryotes :-** But in Eukaryotes some modifications are occure before leaving the nucleus. eg. Caps & tail's are added which helps during journey toward ribosomes. Cap is added at 5th end & Tail is added at 3rd end.

:- A Brief Comparison :-

Prokaryotic Transcription	Eukaryotic transcription.
• occurs in Cytoplasm	• occurs in Nucleus
• No definite phase of occurance	• Occurs in G1 & G2 phases.
• No initiation factors required.	• Initiation factors eg. TFIIA TFIIB are required.
• Polycistronic. polycistronic mRNA in prokaryotes that encodes several proteins.	• Monocistronic. Monocistronic mRNA in Eukaryotes can encodes only one protein.

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Translation :- (CSS, PMS)

- **Dafination :-**

The Synthesis of protein at ribosome's from mRNA is called translation

- **Role of tRNA in Translation :-**

It plays important role in translation. It read's code's on the mRNA & brings different amino Acid's. Thus a polypeptide chain is formed. The tRNA has complimentary pairing of basis.
But tRNA also have some unusal basis These bases interupt normal base pairing & forms loop's in molecule's. Center loop is called Anticodon loop. It has a sequance of 3 unpaired basis called Anticodon. The mRNA codon's form pairing with its complementary anticodon of tRNA. The tRNA carries Amino Acids.

- **Role of Ribosome's in translation :-**

Ribosome's are the sites of translation or protein Synthesis in the cell.
They consist of small & large Subunit's. These Subunits organize the pairing b/w mRNA & tRNA.

- **Ribosomes have 3 main sites :-**

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- **P- Site :-**

It is the site of ribosomes where peptide bonds are formed.

- **A- Site :-**

This site Contains next amino acid with its tRNA.

- **E- Site :-**

It is a site of ribosome where empty t RNA is present.

- **Steps in translation :-**

Translation includes 3 steps

- **Initiation.**

The mRNA binds to a small ribosomal Subunit. the initiation codon on mRNA is (AUG). The initiator Codon of mRNA align's with the (P) site of ribosome's. The larger Subunit's join this Complex. It form's a complete ribosome.

- **Elongation :-**

During elongation stage the ribosome continues to translate each codon in turn. Each Corresponding amino acid is added to the growing chain & linked via bond called peptide bond which is formed by enzyme Peptidyle transferase.
Elongation Continues until all the Codon's are read.

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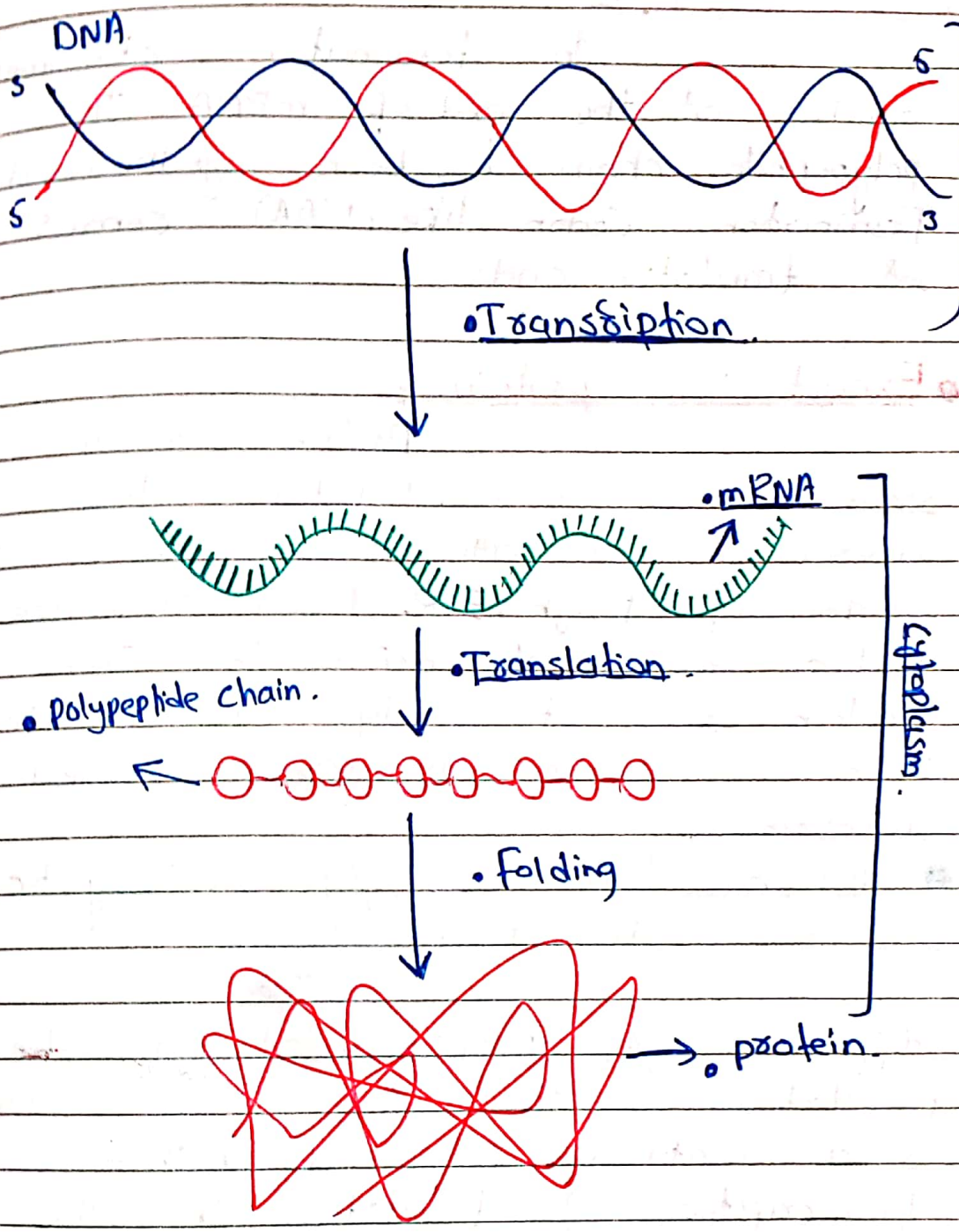

- Do You Know?
  → Translation occurs at the speed of (20 amino Acids) per second.
- Rate of mistakes in translation is 1 per every (100000) nucleotide's. So with our 6 billion base pairs in each Cell about (120000) mistakes occure everytime

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• Termination :-

In termination, ribosomes reaches at the end of mRNA. The polypeptide chain has been synthesized. Termination Codon like (UAA) comes & translation ends.

• Export of protein :-

Protein Synthesis occurs on ribosomes. Most of the ribosomes are attached on the Surface of Rough E.R. Newly synthesized protein moves into E.Reticulum. Then protein moves to Golgi Apparatus where it is packed in vasicles called lysosomes.

• Main difference b/w Prokaryotic & Eukaryotic translation.

The Core difference b/w prokaryoti & Eukaryotic translation is that. prokaryotic translation is a continuous process of both transcription & translation in the Cytoplasm. Bub, Eukaryotic translation is a discontinuous process as translation occurs in Cytoplasm & transcription occurs in Nucleus.

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